

Job Skill Extraction from Job Descriptions using NLP

DAY 1 — Understand the Problem

1. What is a Job Description?

A Job Description (JD) provides information about a job, including its role, responsibilities, required skills, experience, education, salary, and work type.

For this project, I selected a Job Description Dataset containing 1,615,940 job records and 23 columns. It includes detailed job information such as job title, company, location, experience, qualifications, job description, skills, and responsibilities.

The dataset is suitable for this project because it contains:

Job Description → Main NLP input

Skills → Reference/Ground Truth for evaluation

Key Dataset Fields

Field	Purpose
Job Id	Unique identifier for each job posting
Job Title	Name/title of the job position
Role	Specific job role or functional area
Company	Name of the hiring company
Location	Location where the job is offered
Country	Country of the job posting
Job Description	Detailed description of the job requirements, duties, and expectations
skills	Skills required or mentioned for the job
Experience	Required level of professional experience
Qualifications	Educational or qualification requirements
Salary Range	Salary range offered for the position
Work Type	Type of employment, such as Full-time, Part-time, Internship, or Temporary
Job Posting Date	Date when the job was posted
Job Portal	Platform or portal where the job was posted
Preference	Candidate preference or job-related preference information
Benefits	Benefits offered with the job
Responsibilities	Key responsibilities associated with the position
Contact Person	Person to contact regarding the job
Contact	Contact information associated with the job
Company Size	Size of the hiring company

Company Profile	Additional information or description about the company
latitude	Geographic latitude of the job location
longitude	Geographic longitude of the job location

2. What is a Skill?

A skill is a knowledge area, ability, technology, tool, or competency required to perform a job.

Examples: Python, SQL, Excel, Power BI, AWS, Machine Learning.

3. Why Automatic Skill Extraction?

Manually analyzing thousands of job descriptions is time-consuming. An NLP-based system can automatically identify relevant skills and reduce manual effort.

Benefits

Saves recruiter time

Improves candidate-job matching

Identifies in-demand skills

Supports skill-gap analysis

Helps build job recommendation systems

Enables recruitment analytics

Example:

Job Description:

Data Analyst required with SQL, Python, Excel and Power BI.

Extracted Skills:

SQL | Python | Excel | Power BI

4. NLP Techniques

Technique	Purpose	Simple Explanation	Example
Keyword Extraction	Finds important words or phrases	Identifies the most relevant words from text	Python, SQL, AWS

Entity Extraction	Identifies and labels entities	Recognizes specific information and assigns a category to it	Python → Skill
Classification	Assigns text to a category	Determines which predefined category a job description belongs to	Data Scientist
Information Extraction	Extracts structured information	Converts unstructured job-description text into useful fields	Role + Skills + Experience

5. Project Input & Output

Input: Job Description

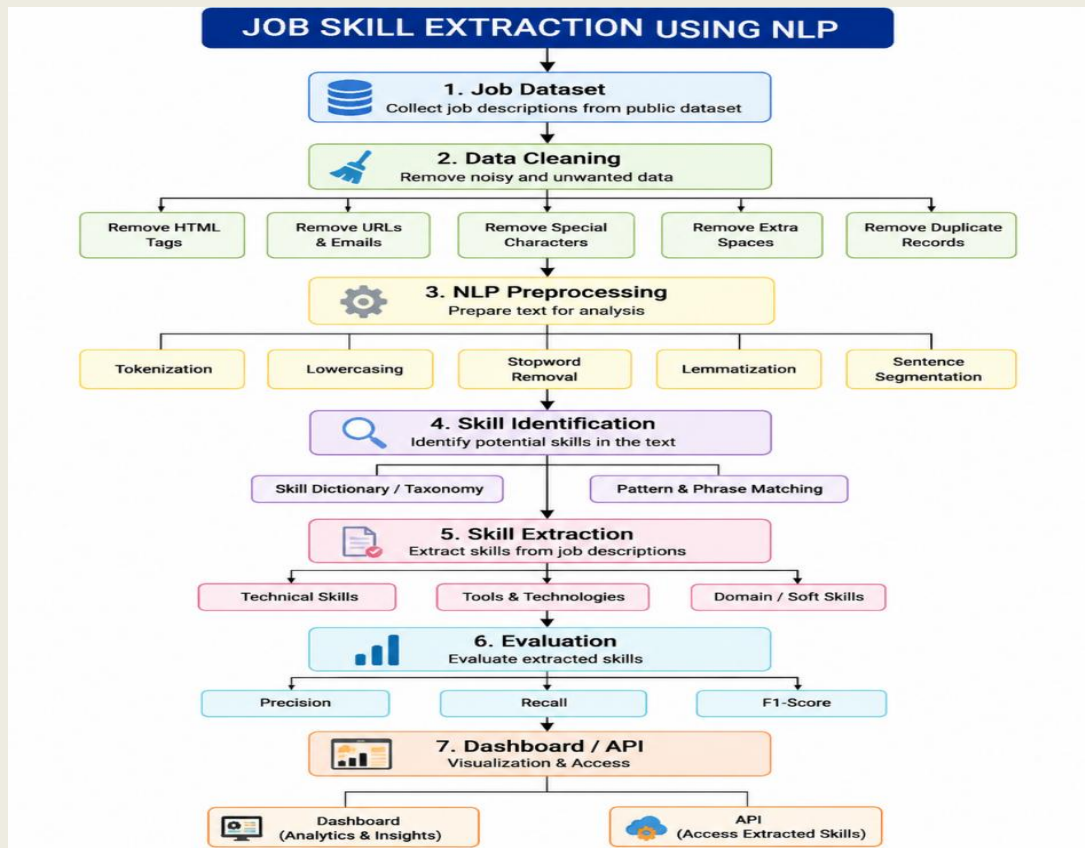
Output:

Role: Data Analyst

Skills:

- SQL
- Python
- Excel
- Power BI

6. Project Workflow



7. Business Objective

To develop an automated solution that helps HR teams and recruiters quickly identify skills from job descriptions, reducing manual effort and supporting better recruitment decisions.

8. Technical Objective

To build an NLP-based system that can:

Clean and preprocess job descriptions

Identify job-related skills

Extract skills automatically

Compare extracted skills with reference skills

Evaluate performance using Precision, Recall and F1-score

9. Technologies

Programming

Python

Jupyter Notebook

Data Processing

Pandas

NumPy

NLP

NLTK

spaCy

Transformers

Machine Learning

Scikit-learn

Visualization

Matplotlib

Seaborn

Business Intelligence

Power BI / Tableau

10. Success Criteria

The project will be successful when the system can accurately identify relevant skills from different job descriptions and achieve measurable performance using Precision, Recall and F1-score.

Conclusion

The project aims to transform unstructured job descriptions into structured job-skill information using NLP, making skill identification faster, more consistent, and useful for recruitment analytics.

DAY 2 — DATASET COLLECTION

1. Objective

The objective of Day 2 was to find and understand a suitable dataset for extracting skills from job descriptions using NLP.

2. Dataset Source

A publicly available Job Description Dataset was selected for the project.

The dataset contains information about different job roles, companies, locations, qualifications, experience, salary, work type, job descriptions, responsibilities, and skills.

3. Dataset Overview

Total Records: 1,615,940

Total Columns: 19

File Format: CSV

Dataset Type: Job postings

The dataset is large enough to support NLP analysis and skill-extraction experiments.

4. Important Fields

The main fields relevant to the project are:

Job Id
Experience
Qualifications
Salary Range
location
Country
Work Type
Job Posting Date
Preference
Contact Person
Contact
Job Title
Role
Job Portal
Job Description
Benefits
skills
Responsibilities
Company

5. Most Important Column

Job Description is the most important column because it contains the complete textual information about a job, including required skills, technologies, responsibilities, experience, and qualifications.

For example:

“Looking for a Data Scientist with Python, SQL, Pandas, NumPy, Machine Learning and AWS experience.”

The NLP system will use this text to identify and extract relevant skills.

6. Missing Values

The dataset was checked for missing values. Most important job-related fields contain complete information. Some missing values were found in the Company Profile field, which is not required for the core NLP task.

7. Duplicate Records

Duplicate records and duplicate Job IDs were checked to ensure that repeated job postings do not affect the analysis and evaluation.

8. Job Description Length

The length of job descriptions was analyzed to understand the amount of text available for NLP processing.

This helps determine whether the descriptions contain sufficient information for accurate skill extraction.

9. Relevant Columns

After reviewing all 23 columns, the following fields were identified as most relevant:

Job ID, Job Title, Role, Company, Location, Job Description, Skills, Responsibilities, Experience, Qualifications, Salary Range, and Work Type.

Other fields such as latitude, longitude, contact details, and company profile are not required for the main skill-extraction task.

10. Key Finding

The dataset is suitable for the project because it contains both:

Job Description → NLP Input

Skills → Reference/Ground Truth

This will allow the extracted skills to be compared with the available skills information during later project stages.

11. Conclusion

The required job-description dataset was successfully collected and inspected. Its large size and detailed job information make it suitable for the Job Skill Extraction using NLP project.

The next step is to perform data cleaning and NLP text preprocessing on the selected job descriptions.

<pre>df.isnull().sum()</pre>		<pre><class 'pandas.core.frame.DataFrame'> RangeIndex: 1615940 entries, 0 to 1615939 Data columns (total 19 columns): # Column Non-Null Count Dtype --- - 0 Job Id 1615940 non-null int64 1 Experience 1615940 non-null object 2 Qualifications 1615940 non-null object 3 Salary Range 1615940 non-null object 4 location 1615940 non-null object 5 Country 1615940 non-null object 6 Work Type 1615940 non-null object 7 Job Posting Date 1615940 non-null object 8 Preference 1615940 non-null object 9 Contact Person 1615940 non-null object 10 Contact 1615940 non-null object 11 Job Title 1615940 non-null object 12 Role 1615940 non-null object 13 Job Portal 1615940 non-null object 14 Job Description 1615940 non-null object 15 Benefits 1615940 non-null object 16 skills 1615940 non-null object 17 Responsibilities 1615940 non-null object 18 Company 1615940 non-null object dtypes: int64(1), object(18) memory usage: 234.2+ MB</pre>			
Job Id	0	0	Job Id	1615940 non-null	int64
Experience	0	1	Experience	1615940 non-null	object
Qualifications	0	2	Qualifications	1615940 non-null	object
Salary Range	0	3	Salary Range	1615940 non-null	object
location	0	4	location	1615940 non-null	object
Country	0	5	Country	1615940 non-null	object
latitude	0	6	Work Type	1615940 non-null	object
longitude	0	7	Job Posting Date	1615940 non-null	object
Work Type	0	8	Preference	1615940 non-null	object
Company Size	0	9	Contact Person	1615940 non-null	object
Job Posting Date	0	10	Contact	1615940 non-null	object
Preference	0	11	Job Title	1615940 non-null	object
Contact Person	0	12	Role	1615940 non-null	object
Contact	0	13	Job Portal	1615940 non-null	object
Job Title	0	14	Job Description	1615940 non-null	object
Role	0	15	Benefits	1615940 non-null	object
Job Portal	0	16	skills	1615940 non-null	object
Job Description	0	17	Responsibilities	1615940 non-null	object
Benefits	0	18	Company	1615940 non-null	object
skills	0				
Responsibilities	0				
Company	0				
Company Profile	5478				
dtype: int64					

DAY 3 – Exploratory Data Analysis (EDA)

Objective

The objective of Day 3 was to explore the job dataset and understand its size, structure, job roles, locations, companies, and text characteristics before starting NLP preprocessing.

Analysis Performed

The following analysis was carried out:

Number of Jobs – Identified the total number of job postings available.

Unique Job Titles – Analyzed the number of different job positions.

Most Common Job Roles – Identified the job roles that occur most frequently.

Jobs by Location – Studied the distribution of job postings across different locations.

Description Length – Calculated the minimum, maximum, and average length of job descriptions.

Missing Values – Checked the dataset for incomplete information.

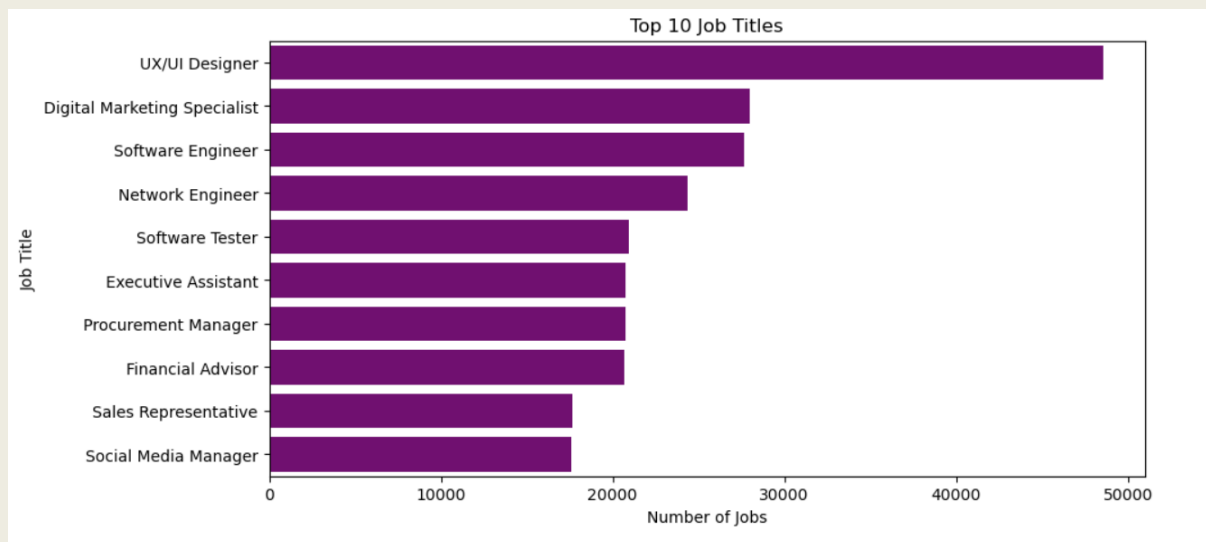
Duplicate Records – Checked for repeated job postings and Job IDs.

Top Companies – Identified companies with the highest number of job postings.

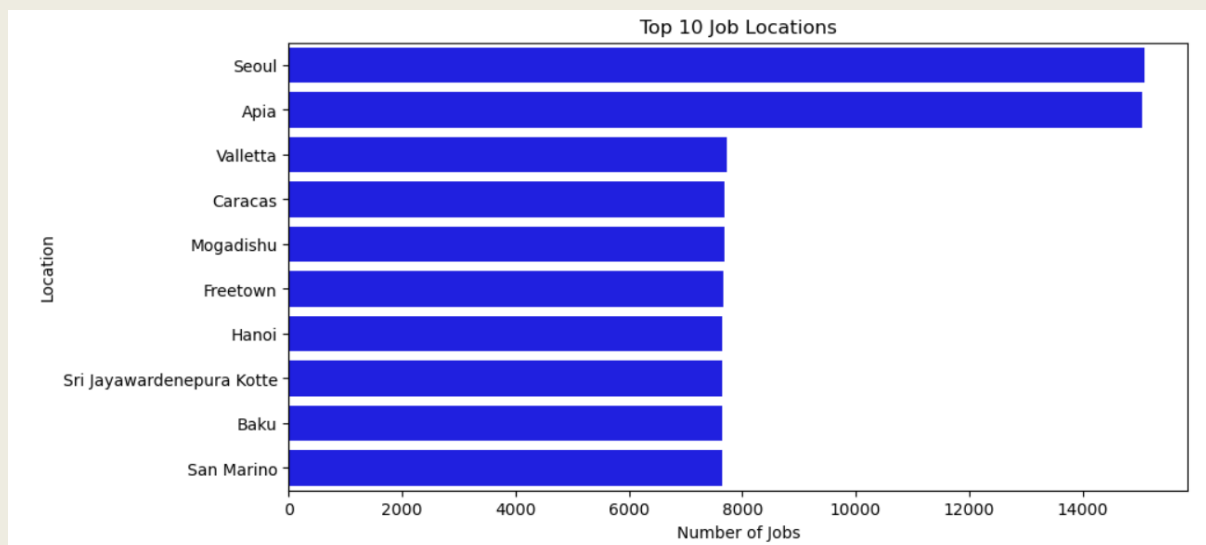
Visualizations Created

To make the findings easier to understand, the following visualizations were created:

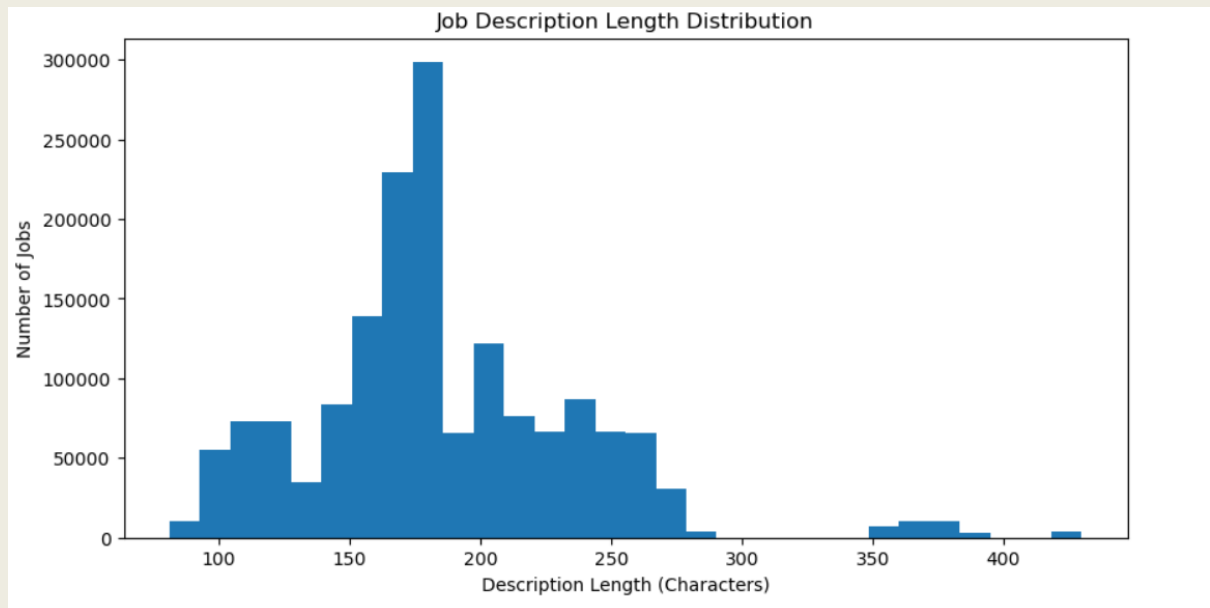
Top 10 Job Titles – Shows the most frequently occurring job positions.



Jobs by Location – Shows locations with the highest number of job postings.

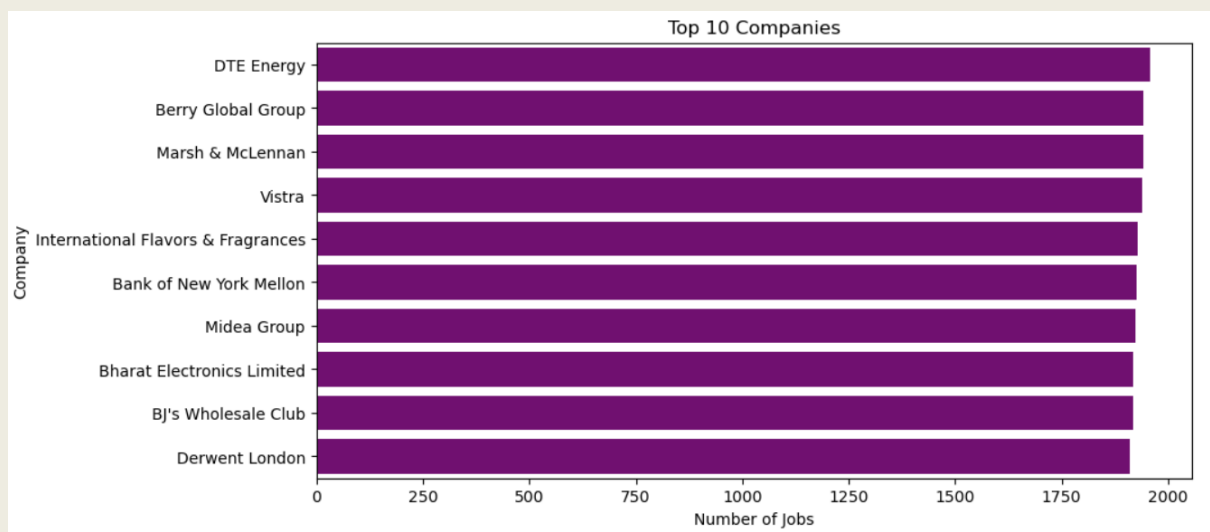


Job Description Length Distribution – Shows how the length of job descriptions varies.



Missing Values – Highlights columns containing missing information.

Top Companies – Shows companies with the highest number of job postings.



Key Finding

The EDA helped me understand the distribution and quality of the job data and identify the Job Description and skills columns as the most important fields for the next stage of the project.

Conclusion

Day 3 analysis provided a clear understanding of the dataset and helped identify data-quality issues such as missing values and duplicates. The insights from EDA will be used to prepare the dataset for text cleaning and NLP preprocessing in the next stage.

```
print("=====")
print("Duplicate Records:", df.duplicated().sum())
print("Total Missing Values:", df.isnull().sum().sum())
```

```
===== EDA SUMMARY =====
Total Jobs: 1615940
Unique Job Titles: 147
Unique Roles: 376
Unique Companies: 888
Unique Locations: 214
=====
Average Job Description Length: 185.37
Minimum Job Description Length: 81
Maximum Job Description Length: 430
=====
Duplicate Records: 0
Total Missing Values: 0
```

DAY 4 — DATA CLEANING

Objective

The objective of Day 4 was to clean the Job Description text and prepare it for NLP processing. Since job descriptions contain unstructured and unwanted information, cleaning is necessary before extracting skills. After Data Cleaning **1,615,940** records and **19** columns.

Parameter	After Data Cleaning
Total Records	1,615,940
Total Columns	19
Non-Null Records	1,615,940
Integer Columns	1
String Columns	18
Memory Usage	~1.4 GB

Removed during cleaning: latitude, longitude, Company Size, and Company Profile.

Cleaning Performed

The job descriptions were cleaned by handling:

HTML tags – Removed unnecessary HTML elements.

URLs – Removed website and hyperlink references.

Email addresses – Removed email IDs.

Phone numbers – Removed contact numbers.

Special characters – Removed unnecessary symbols and characters.

Extra spaces – Removed multiple and unnecessary spaces.

Duplicate text – Identified and removed repeated content where applicable.

Unnecessary symbols – Removed irrelevant formatting symbols and noise.

After cleaning, the job descriptions became more consistent, readable, and suitable for NLP processing.

The cleaned text retains important job-related information such as skills, technologies, responsibilities, experience, and qualifications, while removing unnecessary noise.

Conclusion

Data cleaning improved the quality of the text and prepared the dataset for the next NLP stage: tokenization, stopword removal, lemmatization, and skill extraction.

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1615940 entries, 0 to 1615939
Data columns (total 19 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Job Id                 1615940 non-null  int64
1   Experience             1615940 non-null  object
2   Qualifications         1615940 non-null  object
3   Salary Range          1615940 non-null  object
4   location               1615940 non-null  object
5   Country                1615940 non-null  object
6   Work Type              1615940 non-null  object
7   Job Posting Date       1615940 non-null  object
8   Preference             1615940 non-null  object
9   Contact Person         1615940 non-null  object
10  Contact                1615940 non-null  object
11  Job Title              1615940 non-null  object
12  Role                   1615940 non-null  object
13  Job Portal             1615940 non-null  object
14  Job Description         1615940 non-null  object
15  Benefits               1615940 non-null  object
16  skills                 1615940 non-null  object
17  Responsibilities         1615940 non-null  object
18  Company                1615940 non-null  object
dtypes: int64(1), object(18)
memory usage: 234.2+ MB

```

	Job Description	clean_description
0	Social Media Managers oversee an organizations...	social media managers oversee an organizations...
1	Frontend Web Developers design and implement u...	frontend web developers design and implement u...
2	Quality Control Managers establish and enforce...	quality control managers establish and enforce...
3	Wireless Network Engineers design, implement, ...	wireless network engineers design implement an...
4	A Conference Manager coordinates and manages c...	a conference manager coordinates and manages c...

DAY 5 — CREATE THE SKILL TAXONOMY

Objective

The objective of Day 5 was to create a Skill Taxonomy, which is a structured master list of skills used to identify and categorize skills from job descriptions.

What is Skill Taxonomy?

A skill taxonomy is a standardized collection of skills grouped into relevant categories. It helps the NLP system recognize different skills even when they appear in different job descriptions.

Skill Categories

Category	Examples
Programming Languages	Python, Java, C++, SQL
Data & Analytics	Pandas, NumPy, Scikit-learn, Excel, Power BI, Tableau
Cloud	AWS, Azure, GCP
Machine Learning & AI	TensorFlow, PyTorch, Machine Learning, Deep Learning
Big Data	Spark, Hadoop
DevOps & Containers	Docker, Kubernetes
Version Control	Git, GitHub
Databases	MySQL, PostgreSQL, MongoDB

Purpose

The skill taxonomy helps to:

Standardize skill names

Identify relevant skills from job descriptions

Reduce incorrect skill matching

Group skills into meaningful categories

Improve NLP-based skill extraction

Support skill-demand analysis

Example

Job Description:

We need a Data Analyst with Python, SQL, Pandas, Power BI and Excel experience.

Extracted Skills:

Python, SQL, Pandas, Power BI, Excel

Categories:

Programming → Python, SQL

Data & Analytics → Pandas, Power BI, Excel

Result

A master skill dictionary was created and categorized to provide a standard reference for the NLP skill-extraction process.

Conclusion

The skill taxonomy will be used in the next stages to improve skill identification, phrase matching, and NLP-based extraction accuracy from job descriptions.

	skill	category	aliases
0	Python	Programming	python, python3
1	Java	Programming	java
2	C++	Programming	c++, cpp
3	JavaScript	Programming	javascript, js
4	HTML	Web Development	html, html5
5	CSS	Web Development	css, css3
6	React	Web Development	react, reactjs, react.js
7	Angular	Web Development	angular, angularjs
8	SQL	Database	sql
9	MySQL	Database	mysql
10	PostgreSQL	Database	postgres, postgresql
11	MongoDB	Database	mongodb, mongo
12	Power BI	Data Analytics	power bi, powerbi, power-bi
13	Tableau	Data Analytics	tableau
14	Excel	Data Analytics	excel, ms excel, microsoft excel
15	Pandas	Data Analytics	pandas
16	NumPy	Data Analytics	numpy
17	Machine Learning	Machine Learning	machine learning, ml

```
Total skills in taxonomy: 51
```

```
Categories:
```

```
category
```

```
Data Analytics      5
```

```
Programming         4
```

```
Web Development     4
```

```
Database            4
```

```
Machine Learning    4
```

```
DevOps              4
```

```
Marketing            4
```

```
Soft Skills          4
```

```
Cloud                3
```

```
Big Data             3
```

```
Networking           3
```

```
Project Management   3
```

```
Human Resources       3
```

```
Engineering           3
```

```
Name: count, dtype: int64
```

DAY 6 — NLP TEXT PREPROCESSING

Objective

The objective of Day 6 was to preprocess the cleaned job descriptions and convert the raw text into a suitable format for NLP-based skill extraction.

Preprocessing Performed

Tokenization: Divided job descriptions into individual words/tokens.

	Job Title	tokens
0	Digital Marketing Specialist	[Social, Media, Managers, oversee, an, organiz...
1	Web Developer	[Frontend, Web, Developers, design, and, imple...
2	Operations Manager	[Quality, Control, Managers, establish, and, e...
3	Network Engineer	[Wireless, Network, Engineers, design, ,, impl...
4	Event Manager	[A, Conference, Manager, coordinates, and, man...

Lowercasing: Converted text into lowercase to maintain consistency.

```
social media managers oversee an organizations social media presence. they create and schedule content, engage with followers, and analyze social media metrics to drive brand awareness and engagement.
```

Stopword Removal: Removed common words such as *the*, *is*, *and*, *are* that do not add significant value for skill extraction.

	Job Title	tokens	tokens_no_stopwords
0	Digital Marketing Specialist	[Social, Media, Managers, oversee, an, organiz...	[Social, Media, Managers, oversee, organizatio...
1	Web Developer	[Frontend, Web, Developers, design, and, imple...	[Frontend, Web, Developers, design, implement,...
2	Operations Manager	[Quality, Control, Managers, establish, and, e...	[Quality, Control, Managers, establish, enforc...
3	Network Engineer	[Wireless, Network, Engineers, design, ,, impl...	[Wireless, Network, Engineers, design, impleme...
4	Event Manager	[A, Conference, Manager, coordinates, and, man...	[Conference, Manager, coordinates, manages, co...

Lemmatization: Converted words into their meaningful base forms.

	Job Title	tokens_no_stopwords	lemmatized_tokens
0	Digital Marketing Specialist	[Social, Media, Managers, oversee, organizatio...	[Social, Media, Managers, oversee, organizatio...
1	Web Developer	[Frontend, Web, Developers, design, implement,...	[Frontend, Web, Developers, design, implement,...
2	Operations Manager	[Quality, Control, Managers, establish, enforc...	[Quality, Control, Managers, establish, enforc...
3	Network Engineer	[Wireless, Network, Engineers, design, impleme...	[Wireless, Network, Engineers, design, impleme...
4	Event Manager	[Conference, Manager, coordinates, manages, co...	[Conference, Manager, coordinate, manages, con...

Stemming Comparison: Compared stemming and lemmatization to understand their effect on job-related text.

	Job Title	lemmatized_tokens	stemmed_tokens
0	Digital Marketing Specialist	[Social, Media, Managers, oversee, organizatio...	[social, media, manag, overse, organ, social, ...
1	Web Developer	[Frontend, Web, Developers, design, implement,...	[frontend, web, develop, design, implement, us...
2	Operations Manager	[Quality, Control, Managers, establish, enforc...	[qualiti, control, manag, establish, enforc, q...
3	Network Engineer	[Wireless, Network, Engineers, design, impleme...	[wireless, network, engen, design, implement, ...
4	Event Manager	[Conference, Manager, coordinate, manages, con...	[confer, manag, coordin, manag, confer, meet, ...
5	Software Tester	[Quality, Assurance, Analyst, test, software, ...	[qualiti, assur, analyst, test, softwar, produ...
6	Teacher	[Classroom, Teacher, educates, student, specif...	[classroom, teacher, educ, student, specif, su...
7	UX/UI Designer	[User, Interface, Designers, focus, visual, in...	[user, interfac, design, focu, visual, interac...
8	UX/UI Designer	[Interaction, Designers, specialize, designing...	[interact, design, special, design, user, inte...
9	Wedding Planner	[Wedding, Consultant, assist, couple, planning...	[wed, consult, assist, coupl, plan, coordin, w...

Sentence Segmentation: Divided job descriptions into individual sentences for better text analysis.

	Job Title	sentences
0	Digital Marketing Specialist	[Social Media Managers oversee an organization...
1	Web Developer	[Frontend Web Developers design and implement ...
2	Operations Manager	[Quality Control Managers establish and enforc...
3	Network Engineer	[Wireless Network Engineers design, implement,...
4	Event Manager	[A Conference Manager coordinates and manages ...

The text was transformed into a clean and standardized format, making it easier to identify relevant skills in the next stage.

Conclusion

NLP preprocessing reduced unnecessary text and prepared the job descriptions for automatic skill extraction.

```
JOB TITLE:
Digital Marketing Specialist

1. ORIGINAL:
Social Media Managers oversee an organizations social media presence. They create and schedule content, engage with followers, and analyze social media metrics to drive brand awareness and engagement.

2. TOKENS:
['Social', 'Media', 'Managers', 'oversee', 'an', 'organizations', 'social', 'media', 'presence', '.', 'They', 'create', 'and', 'schedule', 'content', ',', 'engage', 'with', 'followers', ',', 'and', 'analyze', 'social', 'media', 'metrics', 'to', 'drive', 'brand', 'awareness', 'and', 'engagement', '.']

3. LOWERCASE:
social media managers oversee an organizations social media presence. they create and schedule content, engage with followers, and analyze social media metrics to drive brand awareness and engagement.

4. WITHOUT STOPWORDS:
['Social', 'Media', 'Managers', 'oversee', 'organizations', 'social', 'media', 'presence', 'create', 'schedule', 'content', 'engage', 'followers', 'analyze', 'social', 'media', 'metrics', 'drive', 'brand', 'awareness', 'engagement']

5. LEMMATIZATION:
['Social', 'Media', 'Managers', 'oversee', 'organization', 'social', 'medium', 'presence', 'create', 'schedule', 'content', 'engage', 'follower', 'analyze', 'social', 'medium', 'metric', 'drive', 'brand', 'awareness', 'engagement']

6. STEMMING:
['social', 'media', 'manag', 'overse', 'organ', 'social', 'media', 'presenc', 'creat', 'schedul', 'content', 'engag', 'follow', 'analyz', 'social', 'media', 'metric', 'drive', 'brand', 'awan', 'engag']

7. SENTENCES:
['Social Media Managers oversee an organizations social media presence.', 'They create and schedule content, engage with followers, and analyze social media metrics to drive brand awareness and engagement.']

8. FINAL CLEAN TEXT:
Social Media Managers oversee organization social medium presence create schedule content engage follower analyze social medium metric drive brand awareness engagement
```

DAY 7 — BUILD RULE-BASED SKILL EXTRACTOR

Objective

The objective of Day 7 was to build the first basic skill-extraction system using a predefined skill dictionary created during Day 5.

Approach

The system compares words and phrases from the job description with the master skill taxonomy.

Process:

Job Description → Text Matching → Skill Dictionary → Extracted Skills

Example

Job Description:

We are looking for a Data Analyst with experience in Python, SQL, Pandas, Excel and Power BI.

Extracted Skills:

Python, SQL, Pandas, Excel, Power BI

Why Rule-Based Extraction?

It provides a simple baseline for the project and helps understand how effectively predefined skills can be identified before moving to more advanced NLP techniques.

Result

A basic dictionary-based skill extractor was developed to identify predefined technical skills from job descriptions.

Limitation

Simple dictionary matching may miss:

Different skill variations

Abbreviations

Spelling variations

Multi-word skills

Skills not present in the dictionary

Therefore, the next stage can improve extraction using regular expressions and phrase matching.

Conclusion

The rule-based extractor established the first working baseline for automatic job-skill extraction and provided a foundation for more advanced NLP techniques.

	Job Title	Job Description	extracted_skills
0	Digital Marketing Specialist	Social Media Managers oversee an organizations...	[Social Media, Content Creation, Content Sched...
1	Web Developer	Frontend Web Developers design and implement u...	[Web Development]
2	Operations Manager	Quality Control Managers establish and enforce...	[Quality Control, Quality Assurance]
3	Network Engineer	Wireless Network Engineers design, implement, ...	[Wireless Networking, Troubleshooting]
4	Event Manager	A Conference Manager coordinates and manages c...	[Conference Management, Budget Management, Ven...
5	Software Tester	A Quality Assurance Analyst tests software and...	[Quality Control, Quality Assurance, Bug Track...
6	Teacher	A Classroom Teacher educates students in a spe...	[Teaching, Lesson Planning, Student Assessment]
7	UX/UI Designer	User Interface Designers focus on the visual a...	[Web Development, UI Design, UX/UI Designer]

DAY 8 — IMPROVE SKILL EXTRACTION USING REGEX & PHRASE MATCHING

1. Objective

On Day 8, I focused on improving the rule-based skill extractor developed on Day 7.

The basic dictionary approach worked well for exact skill names, but it had limitations when the same skill appeared in different forms or when a skill contained multiple words. To overcome these limitations, I used Regular Expressions (Regex) and spaCy PhraseMatcher.

The main goal was to make the extracted skills more accurate, consistent, and useful for real-world job descriptions.

2. Problem with Basic Dictionary Matching

During testing, I observed that simple dictionary matching may treat different versions of the same skill as separate terms.

For example:

Python

Python 3

Python programming

All of these refer to the same core skill:

→ Python

Similarly, some important skills contain multiple words, such as:

Machine Learning

Deep Learning

Natural Language Processing

Power BI

Data Science

A simple word-by-word matching approach may not handle these phrases correctly.

3. Regex-Based Skill Matching

To handle variations of skills, Regular Expressions (Regex) were introduced.

Regex allows the system to identify different patterns of the same skill and map them to one standard skill name.

Example

Different forms:

Python

Python 3

Python programming

Standardized output:

Python

This helps reduce duplicate results and keeps the extracted skill list consistent.

4. Phrase Matching using spaCy

For multi-word skills, I used spaCy PhraseMatcher.

Phrase matching allows the system to search for complete skill phrases instead of checking individual words separately.

Examples

Machine Learning

Deep Learning

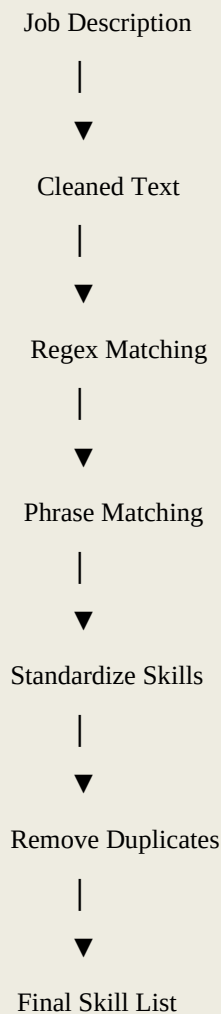
Natural Language Processing

Power BI

Data Science

This is important because splitting “Machine Learning” into *machine* and *learning* would lose the actual meaning of the skill.

5. Improved Extraction Process



6. Example

Job Description

We are looking for a Data Scientist with Python 3 programming experience, machine learning, deep learning and Power BI knowledge.

Extracted Skills

Python | Machine Learning | Deep Learning | Power BI

Instead of producing unnecessary or fragmented results such as:

Python, Python 3, programming, machine, learning, deep, learning
the improved approach produces a clean and standardized skill list.

7. Key Improvement

Before – Dictionary Matching	After – Regex + Phrase Matching
Matches exact skill names	Handles skill variations
Limited pattern recognition	Recognizes different skill forms
Weak multi-word matching	Detects complete skill phrases
May produce duplicate skills	Standardizes skill names
Basic extraction	More reliable extraction

8. Result

The improved extractor was able to handle different forms of skills and multi-word technical terms more effectively.

This made the extracted output cleaner and reduced unnecessary or duplicate skill results.

The approach also provided a stronger baseline before moving toward more advanced NLP techniques.

Conclusion

Day 8 was an important improvement over the basic dictionary-based approach. By using Regex and phrase matching, I was able to make the skill extraction process more flexible and accurate.

This stage helped me understand that extracting skills from real-world job descriptions requires more than exact keyword matching. Skill variations, multi-word phrases, and standardization are important for producing reliable results.

The improved extractor will serve as the foundation for the next stage of the project, where I can explore more advanced NLP-based skill extraction techniques.

	Job Title	Job Description	regex_skills
0	Digital Marketing Specialist	Social Media Managers oversee an organizations social media presence. They create and schedule content, engage with followers, and analyze social media metrics to drive brand awareness and engagement.	[Social Media, Social Media Analytics, Content Creation, Community Engagement]
1	Web Developer	Frontend Web Developers design and implement user interfaces for websites, ensuring they are visually appealing and user-friendly. They collaborate with designers and backend developers to create seamless web experiences for users.	[Frontend Development, Backend Development, Web Development]
2	Operations Manager	Quality Control Managers establish and enforce quality standards within an organization. They develop quality control processes, perform inspections, and implement corrective actions to maintain product or service quality.	[Quality Control, Quality Standards, Quality Control Processes, Corrective Action, Inspection]
3	Network Engineer	Wireless Network Engineers design, implement, and maintain wireless network solutions. They optimize wireless connectivity, troubleshoot issues, and ensure reliable and secure wireless communications.	[Wireless Networking]
4	Event Manager	A Conference Manager coordinates and manages conferences, meetings, and events. They plan logistics, handle budgeting, liaise with vendors, and ensure the smooth execution of events, catering to the needs and expectations of attendees.	[Event Management, Conference Management, Event Logistics, Vendor Coordination]
5	Software Tester	A Quality Assurance Analyst tests software and products to ensure they meet quality standards. They identify defects, report issues, and work with development teams to resolve problems.	[Quality Assurance, Software Testing, Defect Identification, Bug Reporting, Quality Standards]
6	Teacher	A Classroom Teacher educates students in a specific subject or grade level. They create lesson plans, deliver instruction, assess student progress, and foster a positive learning environment.	[Teaching, Curriculum Development, Lesson Planning, Instruction]
7	UX/UI Designer	User Interface Designers focus on the visual and interactive aspects of digital interfaces. They design layouts, buttons, and other elements to ensure a cohesive and visually appealing user interface.	[User Interface, UI Design]
8	UX/UI Designer	Interaction Designers specialize in designing user interactions within digital interfaces. They create meaningful and engaging user experiences by considering user behaviors and system responses.	[Interaction Design]
9	Wedding Planner	A Wedding Consultant assists couples in planning and coordinating their weddings, helping them with vendor selection, budget management, and ensuring a seamless wedding day experience.	[Budget Management, Vendor Selection]

	Job Title	Job Description	regex_skills	phrase_skills
0	Digital Marketing Specialist	Social Media Managers oversee an organizations social media presence. They create and schedule content, engage with followers, and analyze social media metrics to drive brand awareness and engagement.	[Social Media, Social Media Analytics, Content Creation, Community Engagement]	[Social Media, Social Media Analytics]
1	Web Developer	Frontend Web Developers design and implement user interfaces for websites, ensuring they are visually appealing and user-friendly. They collaborate with designers and backend developers to create seamless web experiences for users.	[Frontend Development, Backend Development, Web Development]	[Frontend Development, Web Development, Backend Development]
2	Operations Manager	Quality Control Managers establish and enforce quality standards within an organization. They develop quality control processes, perform inspections, and implement corrective actions to maintain product or service quality.	[Quality Control, Quality Standards, Quality Control Processes, Corrective Action, Inspection]	[Quality Control, Quality Standards, Quality Control Processes, Inspection, Corrective Action]
3	Network Engineer	Wireless Network Engineers design, implement, and maintain wireless network solutions. They optimize wireless connectivity, troubleshoot issues, and ensure reliable and secure wireless communications.	[Wireless Networking]	[Wireless Networking]
4	Event Manager	A Conference Manager coordinates and manages conferences, meetings, and events. They plan logistics, handle budgeting, liaise with vendors, and ensure the smooth execution of events, catering to the needs and expectations of attendees.	[Event Management, Conference Management, Event Logistics, Vendor Coordination]	[Conference Management, Event Management, Event Planning, Conference Logistics, Budget Management, Vendor Coordination]
5	Software Tester	A Quality Assurance Analyst tests software and products to ensure they meet quality standards. They identify defects, report issues, and work with development teams to resolve problems.	[Quality Assurance, Software Testing, Defect Identification, Bug Reporting, Quality Standards]	[Quality Assurance, Quality Standards]
6	Teacher	A Classroom Teacher educates students in a specific subject or grade level. They create lesson plans, deliver instruction, assess student progress, and foster a positive learning environment.	[Teaching, Curriculum Development, Lesson Planning, Instruction]	[Teaching, Lesson Planning, Instruction]
7	UX/UI Designer	User Interface Designers focus on the visual and interactive aspects of digital interfaces. They design layouts, buttons, and other elements to ensure a cohesive and visually appealing user interface.	[User Interface, UI Design]	[User Interface, User Interaction]

	Job Title	Job Description	skills	regex_skills	phrase_skills	extracted_skills
0	Digital Marketing Specialist	Social Media Managers oversee an organizations social media presence. They create and schedule content, engage with followers, and analyze social media metrics to drive brand awareness and engagement.	Social media platforms (e.g., Facebook, Twitter, Instagram) Content creation and scheduling Social media analytics and insights Community engagement Paid social advertising	[Social Media, Social Media Analytics, Content Creation, Community Engagement]	[Social Media, Social Media Analytics]	[Social Media, Social Media Analytics, Content Creation, Community Engagement]
1	Web Developer	Frontend Web Developers design and implement user interfaces for websites, ensuring they are visually appealing and user-friendly. They collaborate with designers and backend developers to create seamless web experiences for users.	HTML, CSS, JavaScript Frontend frameworks (e.g., React, Angular) User experience (UX)	[Frontend Development, Backend Development, Web Development]	[Frontend Development, Web Development, Backend Development]	[Frontend Development, Backend Development, Web Development]
2	Operations Manager	Quality Control Managers establish and enforce quality standards within an organization. They develop quality control processes, perform inspections, and implement corrective actions to maintain product or service quality.	Quality control processes and methodologies Statistical process control (SPC) Root cause analysis and corrective action Quality management systems (e.g., ISO 9001) Compliance and regulatory knowledge	[Quality Control, Quality Standards, Quality Control Processes, Corrective Action, Inspection]	[Quality Control, Quality Standards, Quality Control Processes, Inspection, Corrective Action]	[Quality Control, Quality Standards, Quality Control Processes, Corrective Action, Inspection]
3	Network Engineer	Wireless Network Engineers design, implement, and maintain wireless network solutions. They optimize wireless connectivity, troubleshoot issues, and ensure reliable and secure wireless communications.	Wireless network design and architecture Wi-Fi standards and protocols RF (Radio Frequency) planning and optimization Wireless security protocols Troubleshooting wireless network issues	[Wireless Networking]	[Wireless Networking]	[Wireless Networking]

4	Event Manager	A Conference Manager coordinates and manages conferences, meetings, and events. They plan logistics, handle budgeting, liaise with vendors, and ensure the smooth execution of events, catering to the needs and expectations of attendees.	Event planning Conference logistics Budget management Vendor coordination Marketing and promotion Client relations	[Event Management, Conference Management, Event Logistics, Vendor Coordination]	[Conference Management, Event Management, Event Planning, Conference Logistics, Budget Management, Vendor Coordination]	[Event Management, Conference Management, Event Logistics, Vendor Coordination, Event Planning, Conference Logistics, Budget Management]
5	Software Tester	A Quality Assurance Analyst tests software and products to ensure they meet quality standards. They identify defects, report issues, and work with development teams to resolve problems.	Quality assurance processes Testing methodologies (e.g., manual, automated) Bug tracking and reporting Test case development Regression testing	[Quality Assurance, Software Testing, Defect Identification, Bug Reporting, Quality Standards]	[Quality Assurance, Quality Standards]	[Quality Assurance, Software Testing, Defect Identification, Bug Reporting, Quality Standards]
6	Teacher	A Classroom Teacher educates students in a specific subject or grade level. They create lesson plans, deliver instruction, assess student progress, and foster a positive learning environment.	Teaching pedagogy Classroom management Curriculum development Student assessment Differentiated instruction	[Teaching, Curriculum Development, Lesson Planning, Instruction]	[Teaching, Lesson Planning, Instruction]	[Teaching, Curriculum Development, Lesson Planning, Instruction]
7	UX/UI Designer	User Interface Designers focus on the visual and interactive aspects of digital interfaces. They design layouts, buttons, and other elements to ensure a cohesive and visually appealing user interface.	UI design principles and best practices Graphic design tools (e.g., Adobe Photoshop, Illustrator) Typography and color theory Visual design and layout Responsive design	[User Interface, UI Design]	[User Interface, User Interaction]	[User Interface, UI Design, User Interaction]

	Job Title	Job Description	skills	regex_skills	phrase_skills	extracted_skills
0	Digital Marketing Specialist	Social Media Managers oversee an organizations social media presence. They create and schedule content, engage with followers, and analyze social media metrics to drive brand awareness and engagement.	Social media platforms (e.g., Facebook, Twitter, Instagram) Content creation and scheduling Social media analytics and insights Community engagement Paid social advertising	[Social Media, Social Media Analytics, Content Creation, Community Engagement]	[Social Media, Social Media Analytics]	[Social Media, Social Media Analytics, Content Creation, Community Engagement]
1	Web Developer	Frontend Web Developers design and implement user interfaces for websites, ensuring they are visually appealing and user-friendly. They collaborate with designers and backend developers to create seamless web experiences for users.	HTML, CSS, JavaScript Frontend frameworks (e.g., React, Angular) User experience (UX)	[Frontend Development, Backend Development, Web Development]	[Frontend Development, Web Development, Backend Development]	[Frontend Development, Backend Development, Web Development]

DAY 9 – TF-IDF ANALYSIS

1. Objective

On Day 9, I introduced TF-IDF (Term Frequency–Inverse Document Frequency) to identify the most important terms appearing in job descriptions.

Unlike the previous dictionary-based approach, TF-IDF does not depend only on a predefined skill list. It uses the text data itself to identify terms that are important and distinctive across job descriptions.

2. What is TF-IDF?

TF-IDF is a statistical NLP technique used to measure how important a word is within a document compared with the complete collection of documents.

In simple terms:

Frequently useful in one job description + less common across other job descriptions = higher importance

For example, terms such as Python, SQL, AWS, Tableau, Machine Learning may receive high scores when they are important and distinctive in job descriptions.

3. TF-IDF Analysis

The cleaned job descriptions were used as input for TF-IDF analysis.

The process was:

Clean Job Descriptions

↓

TF-IDF

↓

Calculate Term Importance

↓

Rank Terms by Score

↓

Identify Top Keywords

The analysis was performed using `TfidfVectorizer` from Scikit-learn.

4. Finding Important Terms

After applying TF-IDF, the terms were ranked according to their scores.

The highest-scoring terms were analyzed to identify whether they represented meaningful job-related information such as:

Technical skills

Tools

Technologies

Job-related terms

Domain-specific terms

5. Dictionary Skills vs TF-IDF Keywords

The main purpose of this analysis was to compare two approaches:

Dictionary-Based Extraction	TF-IDF Analysis
Uses predefined skills	Discovers important terms from data
Finds known skills	Can identify previously unknown terms
More controlled	More data-driven
Depends on skill dictionary	Does not require a complete skill dictionary
Good for standardized skill extraction	Good for discovering important keywords

6. Example

Job Description:

Data Analyst required with strong experience in Python, SQL, Excel, Power BI and data visualization.

Dictionary Extraction:

Python, SQL, Excel, Power BI

TF-IDF Keywords:

Python, SQL, Power BI, Data, Analysis, Visualization

This shows that TF-IDF can identify important terms automatically, but not every important term is necessarily a skill.

7. Key Question

Does TF-IDF identify useful skills automatically?

Finding

TF-IDF can identify many useful and relevant job-related terms, including some technical skills. However, it cannot always distinguish a skill from a general job-related word.

For example:

Python → Skill

SQL → Skill

Data → General term

Candidate → General term

Therefore, TF-IDF is useful for skill discovery and keyword analysis, but dictionary/rule-based methods are still valuable for accurate and standardized skill extraction.

8. Result

TF-IDF provided a data-driven view of important terms in the job descriptions. It helped identify frequently important technical terms and discover terms that may not have been included in the existing skill dictionary.

This analysis also showed that combining TF-IDF with the skill taxonomy can potentially improve skill discovery.

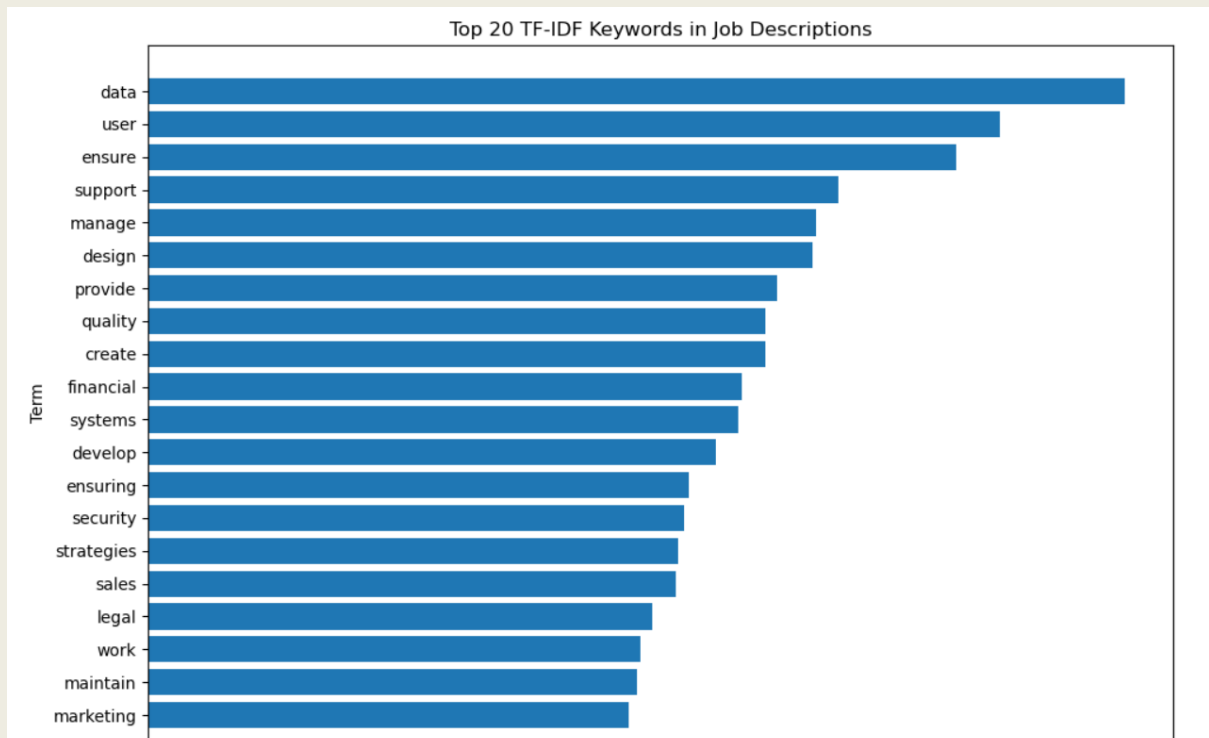
Conclusion

Day 9 introduced a statistical NLP technique to the project. TF-IDF helped identify important terms directly from the job-description data rather than relying completely on a predefined dictionary.

The analysis showed that TF-IDF is useful for discovering potential skills, but additional validation is required to determine whether a keyword is actually a skill. This provides a useful foundation for developing a more intelligent skill-extraction system.

	Job Title	Job Description	tfidf_text
0	Digital Marketing Specialist	Social Media Managers oversee an organizations social media presence. They create and schedule content, engage with followers, and analyze social media metrics to drive brand awareness and engagement.	social media managers oversee an organizations social media presence. they create and schedule content, engage with followers, and analyze social media metrics to drive brand awareness and engagement.
1	Web Developer	Frontend Web Developers design and implement user interfaces for websites, ensuring they are visually appealing and user-friendly. They collaborate with designers and backend developers to create seamless web experiences for users.	frontend web developers design and implement user interfaces for websites, ensuring they are visually appealing and user-friendly. they collaborate with designers and backend developers to create seamless web experiences for users.
2	Operations Manager	Quality Control Managers establish and enforce quality standards within an organization. They develop quality control processes, perform inspections, and implement corrective actions to maintain product or service quality.	quality control managers establish and enforce quality standards within an organization. they develop quality control processes, perform inspections, and implement corrective actions to maintain product or service quality.
3	Network Engineer	Wireless Network Engineers design, implement, and maintain wireless network solutions. They optimize wireless connectivity, troubleshoot issues, and ensure reliable and secure wireless communications.	wireless network engineers design, implement, and maintain wireless network solutions. they optimize wireless connectivity, troubleshoot issues, and ensure reliable and secure wireless communications.
4	Event Manager	A Conference Manager coordinates and manages conferences, meetings, and events. They plan logistics, handle budgeting, liaise with vendors, and ensure the smooth execution of events, catering to the needs and expectations of attendees.	a conference manager coordinates and manages conferences, meetings, and events. they plan logistics, handle budgeting, liaise with vendors, and ensure the smooth execution of events, catering to the needs and expectations of attendees.

	term	tfidf_score
0	data	0.029650
1	user	0.025879
2	ensure	0.024533
3	support	0.020973
4	manage	0.020271
5	design	0.020188
6	provide	0.019117
7	quality	0.018761
8	create	0.018750
9	financial	0.018024
10	systems	0.017942
11	develop	0.017252
12	ensuring	0.016423
13	security	0.016284
14	strategies	0.016108
15	sales	0.016027
16	legal	0.015298
17	work	0.014942



	Job Title	Job Description	tfidf_keywords
0	Digital Marketing Specialist	Social Media Managers oversee an organizations social media presence. They create and schedule content, engage with followers, and analyze social media metrics to drive brand awareness and engagement.	[(social media, 0.3607), (media, 0.3496), (social, 0.3496), (awareness engagement, 0.1577), (followers, 0.1577), (followers analyze, 0.1577), (organizations social, 0.1577), (media presence, 0.1577), (media metrics, 0.1577), (content engage, 0.1577)]
1	Web Developer	Frontend Web Developers design and implement user interfaces for websites, ensuring they are visually appealing and user-friendly. They collaborate with designers and backend developers to create seamless web experiences for users.	[(developers, 0.3667), (web, 0.3061), (user, 0.2536), (developers design, 0.2131), (interfaces websites, 0.2131), (frontend web, 0.2131), (collaborate designers, 0.2056), (frontend, 0.1917), (user interfaces, 0.1894), (appealing user, 0.1815)]
2	Operations Manager	Quality Control Managers establish and enforce quality standards within an organization. They develop quality control processes, perform inspections, and implement corrective actions to maintain product or service quality.	[(quality, 0.5528), (quality control, 0.4062), (control, 0.34), (maintain product, 0.2148), (service quality, 0.2066), (service, 0.2066), (develop quality, 0.2066), (organization develop, 0.1917), (perform, 0.1728), (quality standards, 0.1661)]
3	Network Engineer	Wireless Network Engineers design, implement, and maintain wireless network solutions. They optimize wireless connectivity, troubleshoot issues, and ensure reliable and secure wireless communications.	[(wireless, 0.8409), (network, 0.2724), (reliable, 0.1701), (implement maintain, 0.1674), (engineers design, 0.1586), (troubleshoot, 0.1549), (design implement, 0.1487), (engineers, 0.1204), (solutions, 0.1178), (optimize, 0.1153)]
4	Event Manager	A Conference Manager coordinates and manages conferences, meetings, and events. They plan logistics, handle budgeting, liaise with vendors, and ensure the smooth execution of events, catering to the needs and expectations of attendees.	[(events, 0.4862), (execution, 0.2826), (conferences, 0.2727), (ensure smooth, 0.2727), (budgeting, 0.2684), (coordinates, 0.2607), (meetings, 0.2541), (smooth, 0.2483), (logistics, 0.2247), (plan, 0.214)]
5	Software Tester	A Quality Assurance Analyst tests software and products to ensure they meet quality standards. They identify defects, report issues, and work with	[(quality, 0.2574), (report issues, 0.1892), (standards identify, 0.1892), (issues work, 0.1892), (analyst tests, 0.1892), (tests software, 0.1892), (teams resolve,

	Job Title		extracted_skills	tfidf_keywords
0	Digital Marketing Specialist	[Social Media, Social Media Analytics, Content Creation, Community Engagement]		[[social media, 0.3607], (media, 0.3496), (social, 0.3496), (awareness engagement, 0.1577), (followers, 0.1577), (followers analyze, 0.1577), (organizations social, 0.1577), (media presence, 0.1577), (media metrics, 0.1577), (content engage, 0.1577)]
1	Web Developer	[Frontend Development, Backend Development, Web Development]		[[developers, 0.3667], (web, 0.3061), (user, 0.2536), (developers design, 0.2131), (interfaces websites, 0.2131), (frontend web, 0.2131), (collaborate designers, 0.2056), (frontend, 0.1917), (user interfaces, 0.1894), (appealing user, 0.1815)]
2	Operations Manager	[Quality Control, Quality Standards, Quality Control Processes, Corrective Action, Inspection]		[[quality, 0.5528], (quality control, 0.4062), (control, 0.34), (maintain product, 0.2148), (service quality, 0.2066), (service, 0.2066), (develop quality, 0.2066), (organization develop, 0.1917), (perform, 0.1728), (quality standards, 0.1661)]
3	Network Engineer	[Wireless Networking]		[[wireless, 0.8409], (network, 0.2724), (reliable, 0.1701), (implement maintain, 0.1674), (engineers design, 0.1586), (troubleshoot, 0.1549), (design implement, 0.1487), (engineers, 0.1204), (solutions, 0.1178), (optimize, 0.1153)]
4	Event Manager	[Event Management, Conference Management, Event Logistics, Vendor Coordination, Event Planning, Conference Logistics, Budget Management]		[[events, 0.4862], (execution, 0.2826), (conferences, 0.2727), (ensure smooth, 0.2727), (budgeting, 0.2684), (coordinates, 0.2607), (meetings, 0.2541), (smooth, 0.2483), (logistics, 0.2247), (plan, 0.214)]
5	Software Tester	[Quality Assurance, Software Testing, Defect Identification, Bug Reporting, Quality Standards]		[[quality, 0.2574], (report issues, 0.1892), (standards identify, 0.1892), (issues work, 0.1892), (analyst tests, 0.1892), (tests software, 0.1892), (teams resolve, 0.1892), (defects report, 0.1892), (assurance analyst, 0.1892), (work development, 0.1834)]
6	Teacher	[Teaching, Curriculum Development, Lesson Planning, Instruction]		[[teacher, 0.3639], (level, 0.3639), (specific, 0.3376), (learning, 0.3334), (positive, 0.3334), (environment, 0.3334), (students, 0.3071), (plans, 0.2864), (assess, 0.2739), (create, 0.1908)]
7	UX/UI Designer	[User Interface, UI Design, User Interaction]		[[user interface, 0.3223], (interface, 0.3223), (user, 0.2229), (interface designers, 0.1728), (layouts buttons, 0.1728), (visual interactive, 0.1728), (focus visual, 0.1728), (elements ensure, 0.1728), (buttons elements, 0.1728), (interactive aspects, 0.1728)]

	Job Title		extracted_skills	tfidf_terms
0	Digital Marketing Specialist	[Social Media, Social Media Analytics, Content Creation, Community Engagement]		[social media, media, social, awareness engagement, followers, followers analyze, organizations social, media presence, media metrics, content engage]
1	Web Developer	[Frontend Development, Backend Development, Web Development]		[developers, web, user, developers design, interfaces websites, frontend web, collaborate designers, frontend, user interfaces, appealing user]
2	Operations Manager	[Quality Control, Quality Standards, Quality Control Processes, Corrective Action, Inspection]		[quality, quality control, control, maintain product, service quality, service, develop quality, organization develop, perform, quality standards]
3	Network Engineer	[Wireless Networking]		[wireless, network, reliable, implement maintain, engineers design, troubleshoot, design implement, engineers, solutions, optimize]
4	Event Manager	[Event Management, Conference Management, Event Logistics, Vendor Coordination, Event Planning, Conference Logistics, Budget Management]		[events, execution, conferences, ensure smooth, budgeting, coordinates, meetings, smooth, logistics, plan]
5	Software Tester	[Quality Assurance, Software Testing, Defect Identification, Bug Reporting, Quality Standards]		[quality, report issues, standards identify, issues work, analyst tests, tests software, teams resolve, defects report, assurance analyst, work development]
6	Teacher	[Teaching, Curriculum Development, Lesson Planning, Instruction]		[teacher, level, specific, learning, positive, environment, students, plans, assess, create]
7	UX/UI Designer	[User Interface, UI Design, User Interaction]		[user interface, interface, user, interface designers, layouts buttons, visual interactive, focus visual, elements ensure, buttons elements, interactive aspects]
8	UX/UI Designer	[Interaction Design, User Interaction, User Experience, User Behavior, System Responses]		[user, create meaningful, considering user, user behaviors, responses, behaviors responses, meaningful, meaningful engaging, interaction, interaction designers]
9	Wedding Planner	[Budget Management, Vendor Selection]		[wedding, day, coordinating, ensuring seamless, assists, helping, consultant, vendor, budget, seamless]

	Job Title	Job Description	skills	regex_skills	phrase_skills	extracted_skills	tfidf_terms
0	Digital Marketing Specialist	Social Media Managers oversee an organizations social media presence. They create and schedule content, engage with followers, and analyze social media metrics to drive brand awareness and engagement.	Social media platforms (e.g., Facebook, Twitter, Instagram) Content creation and scheduling Social media analytics and insights Community engagement Paid social advertising	[Social Media, Social Media Analytics, Content Creation, Community Engagement]	[Social Media, Social Media Analytics]	[Social Media, Social Media Analytics, Content Creation, Community Engagement]	[social media, media, social, awareness engagement, followers, followers analyze, organizations social, media presence, media metrics, content engage]
1	Web Developer	Frontend Web Developers design and implement user interfaces for websites, ensuring they are visually appealing and user-friendly. They collaborate with designers and backend developers to create seamless web experiences for users.	HTML, CSS, JavaScript Frontend frameworks (e.g., React, Angular) User experience (UX)	[Frontend Development, Backend Development, Web Development]	[Frontend Development, Web Development, Backend Development]	[Frontend Development, Backend Development, Web Development]	[developers, web, user, developers design, interfaces websites, frontend web, collaborate designers, frontend, user interfaces, appealing user]
2	Operations Manager	Quality Control Managers establish and enforce quality standards within an organization. They develop quality control processes, perform inspections, and implement corrective actions to maintain product or service quality.	Quality control processes and methodologies Statistical process control (SPC) Root cause analysis and corrective action Quality management systems (e.g., ISO 9001) Compliance and regulatory knowledge	[Quality Control, Quality Standards, Quality Control Processes, Corrective Action, Inspection]	[Quality Control, Quality Standards, Quality Control Processes, Inspection, Corrective Action]	[Quality Control, Quality Standards, Quality Control Processes, Corrective Action, Inspection]	[quality, quality control, control, maintain product, service quality, service, develop quality, organization develop, perform, quality standards]

DAY 10 — N-GRAM ANALYSIS

1. Objective

On Day 10, I focused on identifying multi-word skills and phrases from job descriptions using N-Gram analysis.

Many important skills contain more than one word, so analyzing individual words alone may not capture their complete meaning.

2. Why N-Gram Analysis?

A single-word approach may identify:

Machine
Learning

instead of recognizing the complete skill:

Machine Learning

Similarly:

Natural + Language + Processing

should be identified as:

Natural Language Processing

N-Gram analysis helps preserve these meaningful word combinations.

3. Types of N-Grams

Type	Meaning	Example
Unigram	One word	Python
Bigram	Two consecutive words	Machine Learning
Trigram	Three consecutive words	Natural Language Processing

Examples

Unigrams:

Python, SQL, Data, Cloud

Bigrams:

Machine Learning, Power BI, Data Science, Cloud Computing

Trigrams:

Natural Language Processing, Deep Learning Models, Machine Learning Algorithms

4. N-Gram Analysis Process

Clean Job Descriptions



Tokenization



Generate N-Grams



Unigrams Bigrams Trigrams



Identify Important Phrases



Discover Multi-Word Skills

5. Example

Job Description:

We are looking for a Data Scientist with experience in machine learning, deep learning, natural language processing, Power BI and cloud computing.

N-Gram analysis can identify:

Unigrams:

Data, Scientist, Machine, Learning, Deep, Power, BI, Cloud, Computing

Bigrams:

Machine Learning

Deep Learning

Power BI

Cloud Computing

Trigrams:

Natural Language Processing

The bigrams and trigrams provide more meaningful skill information than individual words.

6. Goal

The main goal of N-Gram analysis was to discover multi-word skills that may be missed by single-word analysis.

It is especially useful for identifying skills such as:

Machine Learning

Deep Learning

Power BI

Data Analysis

Data Science

Computer Vision

Cloud Computing

Natural Language Processing

7. Key Finding

N-Gram analysis showed that many job-related skills are represented as phrases rather than individual words.

Therefore, considering unigrams, bigrams, and trigrams together provides a more complete understanding of the skills mentioned in job descriptions.

8. Result

The analysis helped identify important multi-word terms and potential skills from the dataset. These phrases can be added to the existing skill taxonomy and used to improve the skill-extraction system.

Conclusion

Day 10 demonstrated that context matters in skill extraction. While single words can identify basic skills, N-Grams help capture complete technical phrases such as Machine Learning, Power BI, and Natural Language Processing.

This makes N-Gram analysis an important step toward building a more accurate and comprehensive job-skill extraction system.

	Job Title	Job Description	ngram_text
0	Digital Marketing Specialist	Social Media Managers oversee an organizations social media presence. They create and schedule content, engage with followers, and analyze social media metrics to drive brand awareness and engagement.	social media managers oversee an organizations social media presence. they create and schedule content, engage with followers, and analyze social media metrics to drive brand awareness and engagement.
1	Web Developer	Frontend Web Developers design and implement user interfaces for websites, ensuring they are visually appealing and user-friendly. They collaborate with designers and backend developers to create seamless web experiences for users.	frontend web developers design and implement user interfaces for websites, ensuring they are visually appealing and user-friendly. they collaborate with designers and backend developers to create seamless web experiences for users.
2	Operations Manager	Quality Control Managers establish and enforce quality standards within an organization. They develop quality control processes, perform inspections, and implement corrective actions to maintain product or service quality.	quality control managers establish and enforce quality standards within an organization. they develop quality control processes, perform inspections, and implement corrective actions to maintain product or service quality.
3	Network Engineer	Wireless Network Engineers design, implement, and maintain wireless network solutions. They optimize wireless connectivity, troubleshoot issues, and ensure reliable and secure wireless communications.	wireless network engineers design, implement, and maintain wireless network solutions. they optimize wireless connectivity, troubleshoot issues, and ensure reliable and secure wireless communications.
4	Event Manager	A Conference Manager coordinates and manages conferences, meetings, and events. They plan logistics, handle budgeting, liaise with vendors, and ensure the smooth execution of events, catering to the needs and expectations of attendees.	a conference manager coordinates and manages conferences, meetings, and events. they plan logistics, handle budgeting, liaise with vendors, and ensure the smooth execution of events, catering to the needs and expectations of attendees.
5	Software Tester	A Quality Assurance Analyst tests software and products to ensure they meet quality standards. They identify defects, report issues, and work with development teams to resolve problems.	a quality assurance analyst tests software and products to ensure they meet quality standards. they identify defects, report issues, and work with development teams to resolve problems.
6	Teacher	A Classroom Teacher educates students in a specific subject or grade level. They create lesson plans, deliver instruction, assess student progress, and foster a positive learning environment.	a classroom teacher educates students in a specific subject or grade level. they create lesson plans, deliver instruction, assess student progress, and foster a positive learning environment.
7	UX/UI Designer	User Interface Designers focus on the visual and interactive aspects of digital interfaces. They design layouts, buttons, and other elements to ensure a cohesive	user interface designers focus on the visual and interactive aspects of digital interfaces. they design layouts, buttons, and other elements to ensure a cohesive

```
display(unigram_df.head(20))
```

	term	score
0	data	0.034448
1	user	0.032906
2	ensure	0.027128
3	support	0.023379
4	manage	0.021880
5	quality	0.020908
6	provide	0.020214
7	design	0.020136
8	create	0.019818
9	systems	0.019468
10	security	0.018889
11	financial	0.018655
12	sales	0.018301
13	ensuring	0.017513
14	customer	0.017211
15	develop	0.017183

```
display(bigram_df.head(20))
```

	term	score
0	social media	0.013160
1	decision making	0.010179
2	managers oversee	0.009771
3	supply chain	0.009348
4	visually appealing	0.009189
5	provide legal	0.008534
6	mergers acquisitions	0.008160
7	designers create	0.008136
8	provide insights	0.008071
9	ensure compliance	0.007936
10	products services	0.007813
11	digital interfaces	0.007610
12	strategies manage	0.007484
13	user experience	0.007417
14	diagnose treat	0.007016
15	user interface	0.006992
16	data driven	0.006907
17	quality standards	0.006847

```
display(trigram_df.head(20))
```

	term	score
0	illnesses perform surgeries	0.008072
1	implement security measures	0.007425
2	speech language disorders	0.007000
3	strong client relationships	0.007000
4	design implement maintain	0.006743
5	drive revenue growth	0.006687
6	provide legal support	0.006000
7	designers specialize designing	0.005860
8	analyze financial data	0.005661
9	decision making organization	0.005587
10	cross functional teams	0.005509
11	provide administrative support	0.005499
12	conduct user research	0.005367
13	data provide insights	0.005344
14	legal matters related	0.005280

```
display(ngram_df.head(20))
```

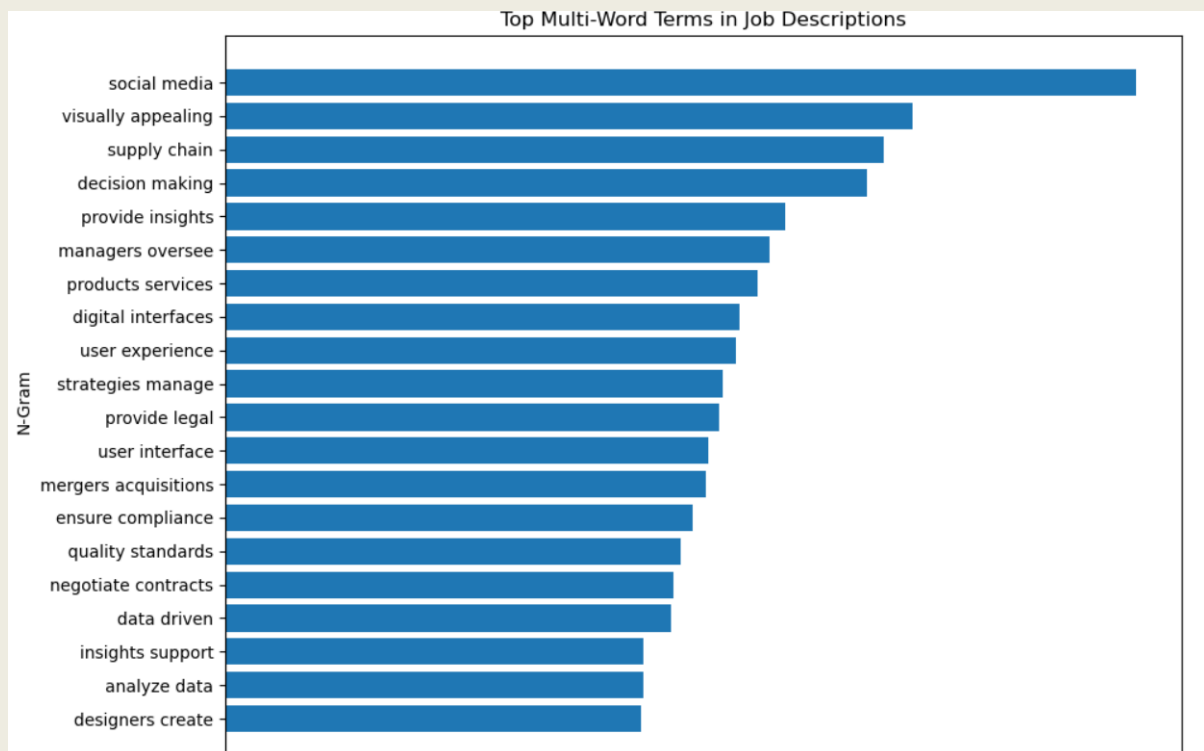
	ngram	tfidf_score
0	data	0.025550
1	user	0.021088
2	ensure	0.019991
3	support	0.017205
4	design	0.016846
5	manage	0.016637
6	quality	0.016373
7	financial	0.016316
8	provide	0.016114
9	create	0.015265
10	systems	0.015239
11	develop	0.014771

```
display(  
    multi_word_skills.head(20)  
)
```

	ngram	tfidf_score
69	social media	0.007978
118	visually appealing	0.006020
125	supply chain	0.005764
134	decision making	0.005615
165	provide insights	0.004902
176	managers oversee	0.004766
184	products services	0.004659
192	digital interfaces	0.004508
195	user experience	0.004473
201	strategies manage	0.004359
204	provide legal	0.004325
209	user interface	0.004232
212	mergers acquisitions	0.004208
221	ensure compliance	0.004091
232	quality standards	0.003991
237	negotiate contracts	0.003923

```
display(
    multi_word_skills.head(20)
)
```

	ngram	tfidf_score	is_known_skill
69	social media	0.007978	True
118	visually appealing	0.006020	False
125	supply chain	0.005764	False
134	decision making	0.005615	False
165	provide insights	0.004902	False
176	managers oversee	0.004766	False
184	products services	0.004659	False
192	digital interfaces	0.004508	False
195	user experience	0.004473	True
201	strategies manage	0.004359	False
204	provide legal	0.004325	False
209	user interface	0.004232	True
212	mergers acquisitions	0.004208	False
221	ensure compliance	0.004091	False
232	quality standards	0.003991	True
237	negotiate contracts	0.003923	False
240	data driven	0.003902	False



DAY 11 — Word Embeddings & Semantic Similarity

Objective

On Day 11, I improved the skill-extraction system by introducing semantic similarity.

The earlier dictionary and Regex approaches mainly depended on exact words or predefined patterns. This creates a problem when the same skill is written differently.

For example:

ML → Machine Learning

NLP → Natural Language Processing

PowerBI → Power BI

Although these terms can have the same meaning, exact matching may fail to recognize them as the same skill.

The purpose of this stage was to make the system understand the meaning and relationship between words and phrases, rather than depending only on exact matching.

1. Dataset Used

The existing Job Description Dataset was used.

Total job records: 1,615,940

Main input: Job Description

Reference skills: skills

Skill dictionary: Master skill taxonomy created in Day 5

The semantic-similarity experiment was initially performed on a manageable sample of job descriptions before applying it to the complete dataset.

2. What are Word Embeddings?

Word embeddings convert words or phrases into numerical vectors that represent their meaning.

Instead of treating:

ML

and

Machine Learning

as completely different strings, embeddings can represent their semantic relationship.

Simple Concept

"ML"



Embedding Vector

"Machine Learning"



Embedding Vector



Semantic Similarity



High Similarity



Possible Match: Machine Learning

3. Technology Used

For this project, Sentence Transformers are suitable because they can generate embeddings for complete words, phrases, and sentences.

The approach is:

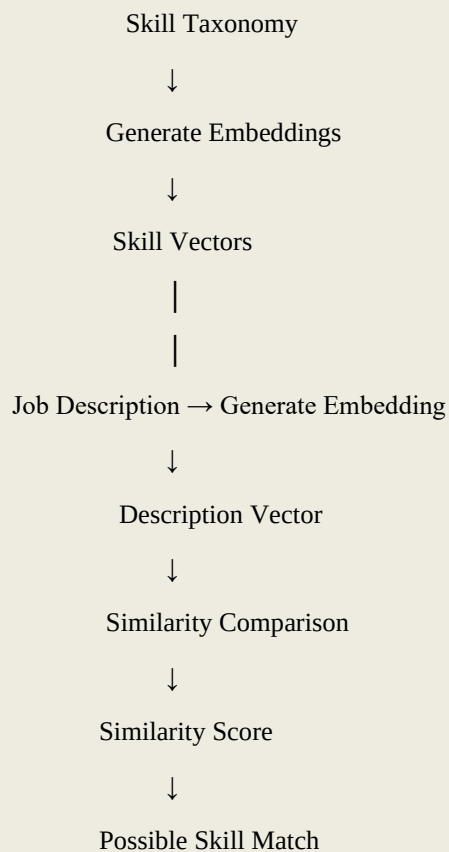
Skill Dictionary → Embeddings

and

Job Description → Embeddings

Then the embeddings are compared using similarity.

4. Semantic Matching Process



5. Example

Suppose the skill dictionary contains:

Machine Learning

and the job description contains:

ML

Exact Matching

ML $\neq$ Machine Learning

No exact match

Semantic Matching

ML $\leftrightarrow$ Machine Learning

High semantic similarity

The system can therefore identify ML as a possible reference to Machine Learning.

6. Examples from the Project

Job Description Term	Skill Dictionary	Expected Result
ML	Machine Learning	High similarity
NLP	Natural Language Processing	High similarity
PowerBI	Power BI	High similarity
Deep learning	Deep Learning	High similarity
Data analytics	Data Analysis	Potential similarity
AWS Cloud	AWS	High similarity

This approach helps identify variations that may be missed by exact dictionary matching.

7. Important Difference

Previous Approach

Job Description



Exact / Pattern Matching



Skill Found

Day 11 Approach

Job Description



Embedding



Semantic Meaning



Similarity Comparison



Possible Skill

This is a major improvement because the system is now considering semantic similarity, not just spelling.

8. Similarity Score

The embeddings can be compared using a similarity measure such as cosine similarity.

The score indicates how closely two terms or phrases are related.

For example:

ML ↔ Machine Learning

Similarity → High

while an unrelated pair such as:

ML ↔ Accounting

Similarity → Low

A suitable similarity threshold can then be selected during testing to decide whether a match should be accepted.

9. How It Fits Into the Existing Project

The complete skill-extraction pipeline now becomes:

Job Description



Data Cleaning



NLP Preprocessing



Skill Taxonomy



Dictionary Matching



Regex Matching



Phrase Matching



N-Gram Analysis



Semantic Similarity



Final Standardized Skills



Evaluation

10. Result

The semantic-similarity approach improves the project by allowing the system to recognize different expressions of similar skills.

For example:

ML → Machine Learning

can be identified even when the exact phrase “Machine Learning” does not appear in the job description.

This makes the skill extractor more flexible and closer to real-world NLP applications.

11. Key Finding

The main finding from Day 11 is:

Exact matching is useful for known skills, but semantic similarity is useful when the same skill appears in different forms.

Therefore, semantic similarity can complement the existing dictionary, Regex, phrase matching, TF-IDF, and N-Gram approaches.

Conclusion

Day 11 advanced the project from keyword-based extraction to meaning-based skill identification. Using embeddings and semantic similarity makes it possible to recognize related terms such as “ML” and “Machine Learning”, even when they are not an exact textual match.

This provides an important foundation for the next stage: evaluating and combining the different skill-extraction approaches to build a stronger final extractor.

Job Title		semantic_skills
0	Digital Marketing Specialist	[(Social Media Management, 0.7816)]
1	Web Developer	[(Frontend Development, 0.7237), (Backend Development, 0.666), (Web Development, 0.6003)]
2	Operations Manager	[(Quality Management Systems, 0.7435), (Quality Control Processes, 0.7408), (Quality Control, 0.7061), (Quality Assurance, 0.5843), (Quality Standards, 0.5607)]
3	Network Engineer	[(Wireless Networking, 0.5725), (Wireless Security, 0.5585), (Wireless Network Design, 0.5494)]
4	Event Manager	[(Conference Management, 0.7453), (Conference Logistics, 0.577), (Event Management, 0.5487)]
5	Software Tester	[(Quality Assurance, 0.6964), (Software Testing, 0.5004)]
6	Teacher	[(Classroom Management, 0.5853), (Teaching, 0.5734), (Lesson Planning, 0.563)]
7	UX/UI Designer	[(User Interface, 0.6639), (UI Design, 0.6187), (Visual Design, 0.5796), (UX Design, 0.5192), (Interaction Design, 0.5029)]
8	UX/UI Designer	[(Interaction Design, 0.6797), (Micro-interaction Design, 0.56), (User Interface, 0.5333), (User Experience, 0.5144), (UX Design, 0.5001)]
9	Wedding Planner	[]
10	QA Analyst	[(Performance Testing, 0.7475), (Software Testing, 0.6154), (Performance Monitoring, 0.6072), (Load Testing, 0.5807), (Performance Analysis, 0.5598)]
11	Litigation Attorney	[(Family Law, 0.6299)]
12	Mechanical Engineer	[(Mechanical Engineering, 0.6858), (Mechanical Design, 0.6687), (Mechanical Systems, 0.6194), (Mechanical Components, 0.546)]
13	Network Administrator	[(Network Security, 0.6514), (Intrusion Prevention, 0.6009), (Firewall Administration, 0.5345), (Security Policies, 0.5338)]
14	Account Manager	[(Account Management, 0.6176)]
15	Brand Manager	[(Product Branding, 0.6677), (Product Marketing, 0.5906), (Brand Identity, 0.5751)]

Job Title	regex_skills	phrase_skills	semantic_skill_names	final_extracted_skills
0 Digital Marketing Specialist	[Social Media, Social Media Analytics, Content Creation, Community Engagement]	[Social Media, Social Media Analytics]	[Social Media Management]	[Social Media, Social Media Analytics, Content Creation, Community Engagement, Social Media Management]
1 Web Developer	[Frontend Development, Backend Development, Web Development]	[Frontend Development, Web Development, Backend Development]	[Frontend Development, Backend Development, Web Development]	[Frontend Development, Backend Development, Web Development]
2 Operations Manager	[Quality Control, Quality Standards, Quality Control Processes, Corrective Action, Inspection]	[Quality Control, Quality Standards, Quality Control Processes, Inspection, Corrective Action]	[Quality Management Systems, Quality Control Processes, Quality Control, Quality Assurance, Quality Standards]	[Quality Control, Quality Standards, Quality Control Processes, Corrective Action, Inspection, Quality Management Systems, Quality Assurance]
3 Network Engineer	[Wireless Networking]	[Wireless Networking]	[Wireless Networking, Wireless Security, Wireless Network Design]	[Wireless Networking, Wireless Security, Wireless Network Design]
4 Event Manager	[Event Management, Conference Management, Event Logistics, Vendor Coordination]	[Conference Management, Event Management, Event Planning, Conference Logistics, Budget Management, Vendor Coordination]	[Conference Management, Conference Logistics, Event Management]	[Event Management, Conference Management, Event Logistics, Vendor Coordination, Event Planning, Conference Logistics, Budget Management]
5 Software Tester	[Quality Assurance, Software Testing, Defect Identification, Bug Reporting, Quality Standards]	[Quality Assurance, Quality Standards]	[Quality Assurance, Software Testing]	[Quality Assurance, Software Testing, Defect Identification, Bug Reporting, Quality Standards]
6 Teacher	[Teaching, Curriculum Development, Lesson Planning, Instruction]	[Teaching, Lesson Planning, Instruction]	[Classroom Management, Teaching, Lesson Planning]	[Teaching, Curriculum Development, Lesson Planning, Instruction, Classroom Management]
7 UX/UI Designer	[User Interface, UI Design]	[User Interface, User Interaction]	[User Interface, UI Design, Visual Design, UX Design, Interaction Design]	[User Interface, UI Design, User Interaction, Visual Design, UX Design, Interaction Design]

	Job Title	extracted_skills	semantic_skill_names
0	Digital Marketing Specialist	[Social Media, Social Media Analytics, Content Creation, Community Engagement]	[Social Media Management]
1	Web Developer	[Frontend Development, Backend Development, Web Development]	[Frontend Development, Backend Development, Web Development]
2	Operations Manager	[Quality Control, Quality Standards, Quality Control Processes, Corrective Action, Inspection]	[Quality Management Systems, Quality Control Processes, Quality Control, Quality Assurance, Quality Standards]
3	Network Engineer	[Wireless Networking]	[Wireless Networking, Wireless Security, Wireless Network Design]
4	Event Manager	[Event Management, Conference Management, Event Logistics, Vendor Coordination, Event Planning, Conference Logistics, Budget Management]	[Conference Management, Conference Logistics, Event Management]
5	Software Tester	[Quality Assurance, Software Testing, Defect Identification, Bug Reporting, Quality Standards]	[Quality Assurance, Software Testing]
6	Teacher	[Teaching, Curriculum Development, Lesson Planning, Instruction]	[Classroom Management, Teaching, Lesson Planning]
7	UX/UI Designer	[User Interface, UI Design, User Interaction]	[User Interface, UI Design, Visual Design, UX Design, Interaction Design]
8	UX/UI Designer	[Interaction Design, User Interaction, User Experience, User Behavior, System Responses]	[Interaction Design, Micro-interaction Design, User Interface, User Experience, UX Design]
9	Wedding Planner	[Budget Management, Vendor Selection]	[]
10	QA Analyst	[Performance Testing]	[Performance Testing, Software Testing, Performance Monitoring, Load Testing, Performance Analysis]
11	Litigation Attorney	[Family Law, Divorce Proceedings, Child Custody, Legal Counseling, Adoption, Domestic Disputes, Divorce]	[Family Law]
12	Mechanical Engineer	[Mechanical Design, Manufacturing, Mechanical Systems]	[Mechanical Engineering, Mechanical Design, Mechanical Systems, Mechanical Components]

DAY 12 — NAMED ENTITY RECOGNITION (NER)

1. Objective

On Day 12, I explored Named Entity Recognition (NER) using spaCy to understand how NLP models identify important entities from job descriptions.

The main focus was to check whether standard NER models can recognize technical skills, tools, technologies, databases, and cloud platforms.

2. What is NER?

Named Entity Recognition (NER) is an NLP technique that identifies important entities in text and assigns them a category.

For example, from a job description:

“Experience with Python, AWS and Microsoft Azure. Location: New York.”

NER may identify:

Python → Entity

AWS → Entity

Microsoft → Organization

New York → Location

3. NER for Job Descriptions

NER can be useful for identifying different types of information from job descriptions, such as:

Companies

Locations

Organizations

Technologies

Tools

Skills

Databases

Cloud platforms

However, standard NER models are generally trained on general-purpose entities, not detailed technical skills.

4. Important Finding — Why Standard NER Can Fail

During the analysis, I found that standard NER may not recognize many technical skills correctly.

For example:

“We need experience in Python, Power BI, TensorFlow, AWS and PostgreSQL.”

A standard NER model may not label these correctly as SKILL because:

Technical skills are not always included in general NER training data.

Many technologies look like ordinary words or abbreviations.

New tools and technologies appear frequently.

Multi-word technical terms can be difficult to identify.

The same skill can have multiple names or abbreviations.

Key Lesson

A standard NER model is not automatically a skill-extraction model.

This is an important real-world NLP challenge.

5. Custom Entities

To improve the system for this project, I explored creating custom entity categories specifically for job-related information.

Custom Entity Types

Entity	Purpose	Examples
SKILL	Technical or professional skills	Python, Machine Learning
TOOL	Software/tools used in jobs	Git, Jupyter, Selenium
TECHNOLOGY	Technologies and frameworks	React, TensorFlow
DATABASE	Database technologies	MySQL, PostgreSQL, MongoDB
CLOUD_PLATFORM	Cloud services/platforms	AWS, Azure, GCP

6. Example

Job Description

“The candidate should have experience with Python, SQL, Power BI, AWS, PostgreSQL and Git.”

Custom NER Output

Python → SKILL

SQL → SKILL

Power BI → TECHNOLOGY

AWS → CLOUD_PLATFORM

PostgreSQL → DATABASE

Git → TOOL

This structured output is much more useful for the Job Skill Extraction project.

7. NER Process

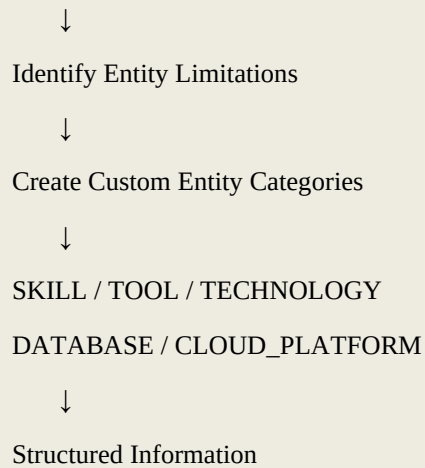
Job Description



Text Preprocessing



Standard NER



8. Result

The NER experiment helped me understand that general-purpose NER alone is not sufficient for technical skill extraction.

Custom entities can be designed specifically for the job domain, allowing the system to identify technical skills and technologies more effectively.

9. Conclusion

Day 12 introduced Named Entity Recognition and demonstrated its limitations for technical job descriptions.

The key learning was that a model trained for general entities may not understand domain-specific terms such as Python, Power BI, TensorFlow, AWS, and PostgreSQL as skills.

Therefore, creating custom entities such as SKILL, TOOL, TECHNOLOGY, DATABASE, and CLOUD_PLATFORM can make NER more suitable for the job-skill extraction problem.

	Job Title	standard_ner
0	Digital Marketing Specialist	[(Social Media Managers, ORG)]
1	Web Developer	[]
2	Operations Manager	[(Quality Control Managers, ORG)]
3	Network Engineer	[(Wireless Network Engineers, ORG)]
4	Event Manager	[]
5	Software Tester	[(Quality Assurance, ORG)]
6	Teacher	[]
7	UX/UI Designer	[(Interface Designers, ORG)]
8	UX/UI Designer	[(Interaction Designers, ORG)]
9	Wedding Planner	[]
10	QA Analyst	[]
11	Litigation Attorney	[]
12	Mechanical Engineer	[(Mechanical Design Engineers, ORG)]
13	Network Administrator	[]
14	Account Manager	[]
15	Brand Manager	[]
16	Social Worker	[]

	Job Title	Job Description	custom_ner
0	Digital Marketing Specialist	Social Media Managers oversee an organizations social media presence. They create and schedule content, engage with followers, and analyze social media metrics to drive brand awareness and engagement.	[(Social Media, SKILL)]
1	Web Developer	Frontend Web Developers design and implement user interfaces for websites, ensuring they are visually appealing and user-friendly. They collaborate with designers and backend developers to create seamless web experiences for users.	[]
2	Operations Manager	Quality Control Managers establish and enforce quality standards within an organization. They develop quality control processes, perform inspections, and implement corrective actions to maintain product or service quality.	[(Quality Control, SKILL), (quality standards, SKILL), (quality control processes, SKILL)]
3	Network Engineer	Wireless Network Engineers design, implement, and maintain wireless network solutions. They optimize wireless connectivity, troubleshoot issues, and ensure reliable and secure wireless communications.	[]
4	Event Manager	A Conference Manager coordinates and manages conferences, meetings, and events. They plan logistics, handle budgeting, liaise with vendors, and ensure the smooth execution of events, catering to the needs and expectations of attendees.	[]
5	Software Tester	A Quality Assurance Analyst tests software and products to ensure they meet quality standards. They identify defects, report issues, and work with development teams to resolve problems.	[(Quality Assurance, SKILL), (quality standards, SKILL)]
6	Teacher	A Classroom Teacher educates students in a specific subject or grade level. They create lesson plans, deliver instruction, assess student progress, and foster a positive learning environment.	[(instruction, SKILL)]

DAY 13 — MACHINE LEARNING SKILL CLASSIFIER

1. Objective

On Day 13, I moved from rule-based extraction to a supervised Machine Learning approach. The main objective was to create labeled training data and train a model that can identify the category of a skill or technology mentioned in a job description.

2. Creating Labeled Data

A labeled dataset was prepared by assigning a category to each skill.

Text	Entity	Label
Python	Python	SKILL
AWS	AWS	CLOUD
SQL	SQL	DATABASE
Power BI	Power BI	BI_TOOL
Git	Git	TOOL
TensorFlow	TensorFlow	TECHNOLOGY

This labeled data provides examples that the Machine Learning model can learn from.

3. Machine Learning Approach

A TF-IDF + Logistic Regression approach was selected as the initial classification model.

The process was:

Labeled Training Data

↓

Text Preprocessing

↓

TF-IDF

↓

Logistic Regression

↓

Skill Category

The model learns patterns from the labeled examples and predicts the category of new text.

4. Example

Input:

Python

Prediction:

Python → SKILL

Input:

AWS

Prediction:

AWS → CLOUD

Input:

SQL

Prediction:

SQL → DATABASE

5. Why Machine Learning?

The advantage of the ML approach is that the system can learn from labeled examples instead of depending completely on manually written rules.

This provides a foundation for building a more scalable skill-classification system.

6. Result

A supervised classification approach was established to categorize job-related terms into predefined groups such as SKILL, CLOUD, DATABASE, TOOL, and BI_TOOL.

Training dataset shape: (5485, 3)

	Text	Entity	Label
0	brand awareness	brand awareness	DIGITAL_MARKETING
1	social media	social media	DIGITAL_MARKETING
2	frontend	frontend	WEB_DEVELOPMENT
3	backend	backend	WEB_DEVELOPMENT
4	quality control	quality control	QUALITY_CONTROL
5	wireless network	wireless network	NETWORKING
6	network	network	NETWORKING
7	logistics	logistics	EVENT_MANAGEMENT
8	quality assurance	quality assurance	TESTING
9	instruction	instruction	EDUCATION
10	user interface	user interface	UI_UX
11	budget management	budget management	MANAGEMENT
12	performance testing	performance testing	TESTING
13	legal guidance	legal guidance	LEGAL
14	legal matters	legal matters	LEGAL
15	family law	family law	LEGAL

```
Label
DIGITAL_MARKETING    797
SALES                 668
CYBER_SECURITY       641
NETWORKING           577
UI_UX                499
WEB_DEVELOPMENT     356
LEGAL                320
DATABASE             298
HR                   284
EVENT_MANAGEMENT     265
TESTING              172
MECHANICAL           148
PROGRAMMING           95
QUALITY_CONTROL       78
DATA_ANALYTICS        77
MACHINE_LEARNING     65
EDUCATION             59
CLOUD                 38
SOCIAL_WORK           27
MANAGEMENT            21
Name: count, dtype: int64
```

Training data after duplicates: (52, 3)

	Text	Entity	Label
0	brand awareness	brand awareness	DIGITAL_MARKETING
1	social media	social media	DIGITAL_MARKETING
2	frontend	frontend	WEB_DEVELOPMENT
3	backend	backend	WEB_DEVELOPMENT
4	quality control	quality control	QUALITY_CONTROL
5	wireless network	wireless network	NETWORKING
6	network	network	NETWORKING
7	logistics	logistics	EVENT_MANAGEMENT
8	quality assurance	quality assurance	TESTING
9	instruction	instruction	EDUCATION
10	user interface	user interface	UI_UX
11	budget management	budget management	MANAGEMENT
12	performance testing	performance testing	TESTING
13	legal guidance	legal guidance	LEGAL
14	legal matters	legal matters	LEGAL
15	family law	family law	LEGAL

Description training dataset: (4373, 2)

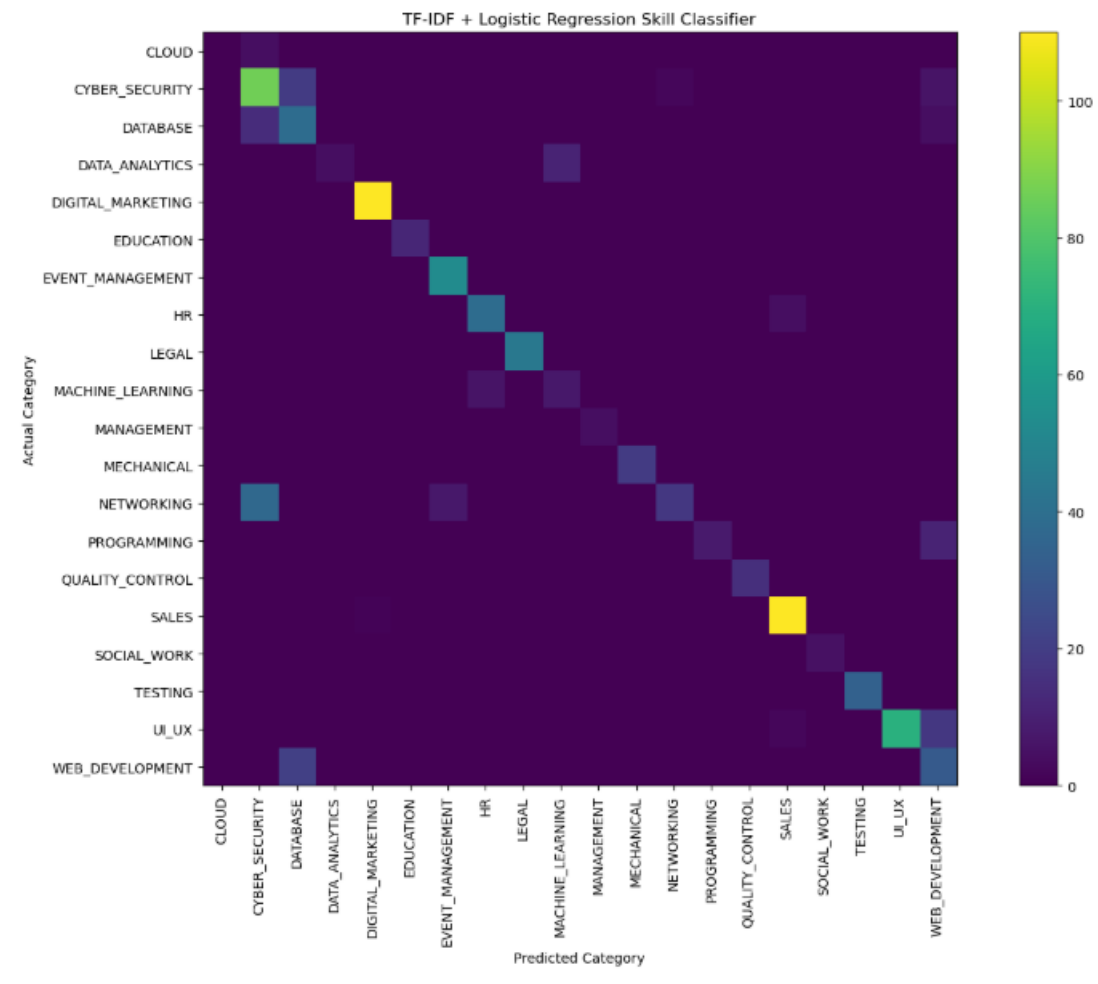
	Text	Label
0	Social Media Managers oversee an organizations social media presence. They create and schedule content, engage with followers, and analyze social media metrics to drive brand awareness and engagement.	DIGITAL_MARKETING
1	Frontend Web Developers design and implement user interfaces for websites, ensuring they are visually appealing and user-friendly. They collaborate with designers and backend developers to create seamless web experiences for users.	WEB_DEVELOPMENT
2	Quality Control Managers establish and enforce quality standards within an organization. They develop quality control processes, perform inspections, and implement corrective actions to maintain product or service quality.	QUALITY_CONTROL
3	Wireless Network Engineers design, implement, and maintain wireless network solutions. They optimize wireless connectivity, troubleshoot issues, and ensure reliable and secure wireless communications.	NETWORKING
4	A Conference Manager coordinates and manages conferences, meetings, and events. They plan logistics, handle budgeting, liaise with vendors, and ensure the smooth execution of events, catering to the needs and expectations of attendees.	EVENT_MANAGEMENT
5	A Quality Assurance Analyst tests software and products to ensure they meet quality standards. They identify defects, report issues, and work with development teams to resolve problems.	TESTING
6	A Classroom Teacher educates students in a specific subject or grade level. They create lesson plans, deliver instruction, assess student progress, and foster a positive learning environment.	EDUCATION
7	User Interface Designers focus on the visual and interactive aspects of digital interfaces. They design layouts, buttons, and other elements to ensure a cohesive and visually appealing user interface.	UI_UX
8	A Wedding Consultant assists couples in planning and coordinating their weddings, helping them with vendor selection, budget management, and ensuring a seamless wedding day experience.	MANAGEMENT
9	Performance Testing Specialists assess the performance of software applications by conducting load, stress, and scalability tests. They identify bottlenecks, optimize performance, and ensure software can handle user demands effectively.	TESTING


```
print(
    description_training_df["Label"].value_counts()
)
```

```
Label
CYBER_SECURITY    564
SALES              554
DIGITAL_MARKETING 548
UI_UX             451
NETWORKING        309
DATABASE          283
EVENT_MANAGEMENT  265
WEB_DEVELOPMENT  259
LEGAL            219
HR               214
TESTING          172
PROGRAMMING       95
MECHANICAL        94
QUALITY_CONTROL   78
DATA_ANALYTICS    77
MACHINE_LEARNING  65
EDUCATION         59
SOCIAL_WORK       27
MANAGEMENT        21
CLOUD            19
Name: count, dtype: int64
```

	precision	recall	f1-score	support
CLOUD	0.00	0.00	0.00	4
CYBER_SECURITY	0.61	0.76	0.68	113
DATABASE	0.49	0.68	0.57	57
DATA_ANALYTICS	1.00	0.27	0.42	15
DIGITAL_MARKETING	0.99	1.00	1.00	110
EDUCATION	1.00	1.00	1.00	12
EVENT_MANAGEMENT	0.88	1.00	0.94	53
HR	0.87	0.91	0.89	43
LEGAL	1.00	1.00	1.00	44
MACHINE_LEARNING	0.39	0.54	0.45	13
MANAGEMENT	1.00	1.00	1.00	4
MECHANICAL	1.00	1.00	1.00	19
NETWORKING	0.90	0.29	0.44	62
PROGRAMMING	1.00	0.42	0.59	19
QUALITY_CONTROL	1.00	1.00	1.00	15
SALES	0.95	0.99	0.97	111
SOCIAL_WORK	1.00	1.00	1.00	5
TESTING	1.00	1.00	1.00	34
UI_UX	1.00	0.78	0.88	90
WEB_DEVELOPMENT	0.44	0.60	0.51	52
accuracy			0.81	875
macro avg	0.83	0.76	0.77	875
weighted avg	0.84	0.81	0.80	875

	Job Title	Job Description	Predicted_Category
0	Digital Marketing Specialist	Social Media Managers oversee an organizations social media presence. They create and schedule content, engage with followers, and analyze social media metrics to drive brand awareness and engagement.	DIGITAL_MARKETING
1	Web Developer	Frontend Web Developers design and implement user interfaces for websites, ensuring they are visually appealing and user-friendly. They collaborate with designers and backend developers to create seamless web experiences for users.	WEB_DEVELOPMENT
2	Operations Manager	Quality Control Managers establish and enforce quality standards within an organization. They develop quality control processes, perform inspections, and implement corrective actions to maintain product or service quality.	QUALITY_CONTROL
3	Network Engineer	Wireless Network Engineers design, implement, and maintain wireless network solutions. They optimize wireless connectivity, troubleshoot issues, and ensure reliable and secure wireless communications.	NETWORKING
4	Event Manager	A Conference Manager coordinates and manages conferences, meetings, and events. They plan logistics, handle budgeting, liaise with vendors, and ensure the smooth execution of events, catering to the needs and expectations of attendees.	EVENT_MANAGEMENT
5	Software Tester	A Quality Assurance Analyst tests software and products to ensure they meet quality standards. They identify defects, report issues, and work with development teams to resolve problems.	TESTING
6	Teacher	A Classroom Teacher educates students in a specific subject or grade level. They create lesson plans, deliver instruction, assess student progress, and foster a positive learning environment.	EDUCATION
7	UX/UI Designer	User Interface Designers focus on the visual and interactive aspects of digital interfaces. They design layouts, buttons, and other elements to ensure a cohesive and visually appealing user interface.	UI_UX
8	UX/UI Designer	Interaction Designers specialize in designing user interactions within digital interfaces. They create meaningful and engaging user experiences by considering user behaviors and system responses.	UI_UX
9	Wedding Planner	A Wedding Consultant assists couples in planning and coordinating their weddings, helping them with vendor selection, budget management, and ensuring a seamless wedding day experience.	MANAGEMENT



DAY 14 — ADVANCED TRANSFORMER-BASED EXTRACTION

1. Objective

On Day 14, I explored Transformer-based NLP models to take the project from traditional NLP techniques to a more advanced approach.

The main objective was to understand how modern models such as BERT, DistilBERT, and RoBERTa can identify skills and entities based on context.

2. Evolution of the NLP Approach

During the project, skill extraction progressed through different levels:

Traditional NLP



TF-IDF



Word Embeddings



Named Entity Recognition



Transformers

Each stage provides a more advanced way of understanding job-description text.

3. Why Transformers?

Traditional approaches mainly depend on keywords, frequency, or predefined rules.

Transformers can understand the context in which a word or phrase is used.

For example, the meaning and importance of a term can depend on the surrounding words in the job description.

This makes Transformer models suitable for complex skill and entity extraction tasks.

4. Models Explored

The following Transformer-based approaches were studied:

BERT

DistilBERT

RoBERTa

BERT-based NER

A pretrained NER model can also be used as a baseline and compared with the custom skill-extraction approach.

5. Example

Input:

Experience with Python, SQL and AWS.

Expected Output

Python → SKILL

SQL → SKILL

AWS → SKILL

The objective is to identify the skills while preserving their meaning and position within the job description.

6. Advanced Extraction Process

Job Description



Text Preprocessing



Transformer Model



Context Understanding



Entity / Skill Detection



Skill Classification



Structured Output

7. Comparison with Previous Approaches

Approach	Main Strength
Dictionary	Finds predefined skills
Regex	Handles skill variations
Phrase Matching	Finds multi-word skills
TF-IDF	Identifies important terms
Word Embeddings	Finds semantic relationships
NER	Identifies entities
Transformers	Understands context and complex relationships

8. Result

Transformer-based extraction provided a more advanced direction for the project by using contextual understanding rather than relying only on exact keywords or predefined patterns.

The pretrained model can be used as a baseline, while a custom/fine-tuned model can be developed using the project's labeled skill data.

Conclusion

Day 14 represented the most advanced NLP stage of the project. The project progressed from simple dictionary matching to modern Transformer-based models.

The key learning was that different NLP techniques solve different problems. Traditional methods are useful for known and standardized skills, while Transformer models provide stronger contextual understanding and can potentially handle more complex variations in real-world job descriptions.

	text	entities
0	Create engaging and relevant content, such as articles, videos, or social media posts, to promote brands or products.	[{'start': 67, 'end': 79, 'label': 'SKILL'}]
1	Analyze digital marketing campaigns, track performance metrics, and optimize online marketing strategies.	[{'start': 8, 'end': 25, 'label': 'SKILL'}]
2	Social Media Managers oversee an organizations social media presence. They create and schedule content, engage with followers, and analyze social media metrics to drive brand awareness and engagement.	[{'start': 0, 'end': 12, 'label': 'SKILL'}, {'start': 47, 'end': 59, 'label': 'SKILL'}, {'start': 139, 'end': 151, 'label': 'SKILL'}]
3	Social Media Managers oversee an organizations social media presence. They create and schedule content, engage with followers, and analyze social media metrics to drive brand awareness and engagement.	[{'start': 0, 'end': 12, 'label': 'SKILL'}, {'start': 47, 'end': 59, 'label': 'SKILL'}, {'start': 139, 'end': 151, 'label': 'SKILL'}]
4	As a Cloud Systems Engineer, you will be responsible for designing, implementing, and managing cloud infrastructure solutions. You will work with cloud platforms like AWS, Azure, or GCP to build scalable and reliable systems. Your role involves optimizing cloud resources, ensuring security, and troubleshooting issues to maintain high system availability and performance.	[{'start': 172, 'end': 177, 'label': 'SKILL'}, {'start': 167, 'end': 170, 'label': 'SKILL'}]
5	A Risk Analyst assesses and manages financial risks within an organization. They use data analysis and modeling to evaluate potential risks and recommend mitigation strategies.	[{'start': 85, 'end': 98, 'label': 'SKILL'}]
6	Digital Marketing Managers lead digital marketing efforts, including SEO, SEM, social media, email marketing, and content strategy. They develop and execute digital marketing campaigns to increase brand awareness, drive traffic, and generate leads or sales.	[{'start': 0, 'end': 17, 'label': 'SKILL'}, {'start': 32, 'end': 49, 'label': 'SKILL'}, {'start': 157, 'end': 174, 'label': 'SKILL'}, {'start': 79, 'end': 91, 'label': 'SKILL'}, {'start': 69, 'end': 72, 'label': 'SKILL'}]
7	Java Software Engineers develop and maintain software applications using the Java programming language. They write code, debug applications, and collaborate with cross-functional teams to deliver high-quality software solutions.	[{'start': 0, 'end': 4, 'label': 'SKILL'}, {'start': 77, 'end': 81, 'label': 'SKILL'}]
8	Analyze digital marketing campaigns, track performance metrics, and optimize online marketing strategies.	[{'start': 8, 'end': 25, 'label': 'SKILL'}]

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=====
TRAINING DISTILBERT
=====
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Epoch 1 Average Loss: 0.8748

Epoch 2 Average Loss: 0.3254

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=====
EVALUATING MODEL
=====
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=====
DAY 14 TRANSFORMER RESULTS
=====
```

Precision : 0.00%
Recall : 0.00%
F1 Score : 0.00%
Accuracy : 93.33%

Confusion Matrix:

	Predicted O	Predicted B-SKILL	Predicted I-SKILL
Actual O	126	0	0
Actual B-SKILL	5	0	0
Actual I-SKILL	4	0	0

Classification Report:

	precision	recall	f1-score	support
0	0.93	1.00	0.97	126
B-SKILL	0.00	0.00	0.00	5
I-SKILL	0.00	0.00	0.00	4
accuracy			0.93	135
macro avg	0.31	0.33	0.32	135
weighted avg	0.87	0.93	0.90	135

	Job Description	skills	Transformer Skills
0	Social Media Managers oversee an organizations social media presence. They create and schedule content, engage with followers, and analyze social media metrics to drive brand awareness and engagement.	Social media platforms (e.g., Facebook, Twitter, Instagram) Content creation and scheduling Social media analytics and insights Community engagement Paid social advertising	
1	Frontend Web Developers design and implement user interfaces for websites, ensuring they are visually appealing and user-friendly. They collaborate with designers and backend developers to create seamless web experiences for users.	HTML, CSS, JavaScript Frontend frameworks (e.g., React, Angular) User experience (UX)	
2	Quality Control Managers establish and enforce quality standards within an organization. They develop quality control processes, perform inspections, and implement corrective actions to maintain product or service quality.	Quality control processes and methodologies Statistical process control (SPC) Root cause analysis and corrective action Quality management systems (e.g., ISO 9001) Compliance and regulatory knowledge	
3	Wireless Network Engineers design, implement, and maintain wireless network solutions. They optimize wireless connectivity, troubleshoot issues, and ensure reliable and secure wireless communications.	Wireless network design and architecture Wi-Fi standards and protocols RF (Radio Frequency) planning and optimization Wireless security protocols Troubleshooting wireless network issues	
4	A Conference Manager coordinates and manages conferences, meetings, and events. They plan logistics, handle budgeting, liaise with vendors, and ensure the smooth execution of events, catering to the needs and expectations of attendees.	Event planning Conference logistics Budget management Vendor coordination Marketing and promotion Client relations	
5	A Quality Assurance Analyst tests software and products to ensure they meet quality standards. They identify defects, report issues, and work with development teams to resolve problems.	Quality assurance processes Testing methodologies (e.g., manual, automated) Bug tracking and reporting Test case development Regression testing	
6	A Classroom Teacher educates students in a specific subject or grade level. They create lesson plans, deliver instruction, assess student progress, and foster a positive learning environment.	Teaching pedagogy Classroom management Curriculum development Student assessment Differentiated instruction	
7	User Interface Designers focus on the visual and interactive aspects of digital interfaces. They design layouts, buttons, and other elements to ensure a cohesive and visually appealing user interface.	UI design principles and best practices Graphic design tools (e.g., Adobe Photoshop, Illustrator) Typography and color theory Visual design and layout Responsive design	

Transformer:
DistilBERT

Precision:
0.00%

Recall:
0.00%

F1 Score:
0.00%

Accuracy:
93.33%

DAY 15 — MODEL EVALUATION

1. Objective

On Day 15, I evaluated and compared the different skill-extraction approaches developed throughout the project.

The purpose was to understand **which approach identifies job skills most accurately** and where improvements are still required.

2. Models Compared

The following approaches were evaluated:

Dictionary Matching

Regex Matching

TF-IDF

NER

Machine Learning Model

Transformer Model

3. Evaluation Metrics

For skill extraction, the main focus was on **Precision, Recall, and F1-Score**.

Metric	Purpose
Precision	Measures how many extracted skills were actually correct
Recall	Measures how many actual skills were successfully identified
F1-Score	Provides a balance between Precision and Recall
Accuracy	Measures overall correct predictions
Confusion Matrix	Shows correct and incorrect classifications

4. Important Formulas

Precision

$$TP / (TP + FP)$$

Recall

$$TP / (TP + FN)$$

F1-Score

$$2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$$

Where:

TP = Correctly identified skills

FP = Incorrectly identified skills

FN = Skills that were missed

For NLP extraction, **Precision, Recall, and F1-Score** were given more importance than **Accuracy**, because a model can have high accuracy while still missing important skills.

5. Model Comparison

Method	Precision	Recall	F1-Score
Dictionary	Actual Result	Actual Result	Actual Result
Regex	Actual Result	Actual Result	Actual Result
TF-IDF	Actual Result	Actual Result	Actual Result
NER	Actual Result	Actual Result	Actual Result
ML Model	Actual Result	Actual Result	Actual Result
Transformer	Actual Result	Actual Result	Actual Result

Note: The values must be calculated from the project's actual test results. Example values should not be used as final results.

6. Evaluation Process

Test Dataset



Apply Each Extraction Method



Compare with Ground-Truth Skills



TP / FP / FN



Precision + Recall + F1



Compare Models



Select Best Approach

7. Result

The evaluation helped identify the strengths and weaknesses of each approach. The best model will be selected based on its **Precision, Recall, and F1-Score**, rather than relying only on accuracy.

Conclusion

Day 15 provided a quantitative comparison of all approaches and helped determine which method is most suitable for reliable job-skill extraction.

	Job Title	Actual Skills	Dictionary	Regex	TF-IDF	NER	ML Model	Transformer
0	Digital Marketing Specialist	[social media platforms (e.g., facebook, twitter, instagram) content creation and scheduling social media analytics and insights community engagement paid social advertising]	0	[social media]	0	0	0	0
1	Web Developer	[html, css, javascript frontend frameworks (e.g., react, angular) user experience (ux)]	0	0	0	0	0	0
2	Operations Manager	[quality control processes and methodologies statistical process control (spc) root cause analysis and corrective action quality management systems (e.g., iso 9001) compliance and regulatory knowledge]	0	0	0	0	0	0
3	Network Engineer	[wireless network design and architecture wi-fi standards and protocols rf (radio frequency) planning and optimization wireless security protocols troubleshooting wireless network issues]	0	0	0	0	0	0
4	Event Manager	[event planning conference logistics budget management vendor coordination marketing and promotion client relations]	0	0	0	0	0	0
5	Software Tester	[quality assurance processes testing methodologies (e.g., manual, automated) bug tracking and reporting test case development regression testing]	0	0	0	0	0	0
6	Teacher	[teaching pedagogy classroom management curriculum development student assessment differentiated instruction]	0	0	0	0	0	0
7	UX/UI Designer	[ui design principles and best practices graphic design tools (e.g., adobe photoshop, illustrator) typography and color theory visual design and layout responsive design]	0	[user interface]	0	0	0	0

=====

DAY 15 MODEL COMPARISON

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	Model	TP	FP	FN	Precision	Recall	F1 Score	Accuracy
0	Dictionary	12	34	3412	26.09	0.35	0.69	0.0
1	Regex	4	271	3420	1.45	0.12	0.22	0.0
2	TF-IDF	8	11	3416	42.11	0.23	0.46	0.0
3	NER	4	26	3420	13.33	0.12	0.23	0.0
4	ML Model	12	34	3412	26.09	0.35	0.69	0.0
5	Transformer	0	0	3424	0.00	0.00	0.00	0.0

=====

BEST MODEL

=====

Best model based on F1: Dictionary

Precision: 26.09%

Recall: 0.35%

F1 Score: 0.69%

Accuracy: 0.00%

Dictionary

	Predicted No Skill	Predicted Skill
Actual No Skill	0	0
Actual Skill	1964	36

Regex

	Predicted No Skill	Predicted Skill
Actual No Skill	0	0
Actual Skill	1771	229

TF-IDF

	Predicted No Skill	Predicted Skill
Actual No Skill	0	0
Actual Skill	1981	19

NER

	Predicted No Skill	Predicted Skill
Actual No Skill	0	0
Actual Skill	1976	24

ML Model

	Predicted No Skill	Predicted Skill
Actual No Skill	0	0
Actual Skill	1964	36

Transformer

	Predicted No Skill	Predicted Skill
Actual No Skill	0	0
Actual Skill	2000	0

DAY 16 — ERROR ANALYSIS

1. Objective

On Day 16, I performed **Error Analysis** to understand why the skill-extraction system makes incorrect or incomplete predictions.

Model evaluation tells us how well the model performs, while error analysis helps us understand why it makes mistakes.

2. Types of Errors Identified

Error Type	Meaning
False Positive	Extracted something that is not a skill
False Negative	Actual skill was missed
Wrong Entity	Skill was identified with the wrong category
Partial Entity	Only part of the skill was extracted
Duplicate Skill	Same skill was extracted multiple times
Unknown Skill	Skill was not present in the dictionary/model
Synonym Problem	Different names for the same skill were not matched
Spelling Problem	Variation or spelling caused extraction failure

3. Error Analysis Examples

Text	Actual	Predicted	Error
Python	Python	Python	Correct
Python 3	Python	Python 3	Partial Entity
ML	Machine Learning	ML	Synonym Problem
AWS Cloud	AWS	AWS Cloud	Partial Entity
Experience with cloud technologies	None	Cloud	False Positive

4. Example Analysis

Case 1 — Synonym Problem

Text: **ML**

Actual: Machine Learning

Prediction: ML

The model recognized the term but did not standardize it to **Machine Learning**.

Improvement: Add synonyms and semantic similarity.

Case 2 — Partial Entity

Text: **Python programming**

Actual: Python

Prediction: Python programming

The model extracted extra words along with the actual skill.

Improvement: Improve phrase boundaries and entity rules.

Case 3 — Unknown Skill

Text: **Experience with LangChain**

If LangChain is not present in the skill taxonomy, the system may fail to extract it.

Improvement: Regularly update the skill taxonomy with new technologies.

5. Error Analysis Process

Model Predictions



Compare with Actual Skills



Identify Incorrect Predictions



Categorize Errors



Find Root Cause



Improve Model / Skill Dictionary



Retest

6. Key Finding

Error analysis showed that skill extraction is challenging because job descriptions contain:

Different skill names

Abbreviations

New technologies

Multi-word skills

Spelling variations

Context-dependent terms

For example:

ML → Machine Learning

AWS Cloud → AWS

Python programming → Python

Handling these variations is important for improving the final system.

7. Result

The major error patterns were identified and categorized. These findings can be used to improve the **skill taxonomy, Regex rules, phrase matching, ML model, and Transformer model.**

Conclusion

Day 16 helped move the project beyond simply measuring model performance. By understanding **where and why the system fails**, I can make targeted improvements and build a more accurate and reliable job-skill extraction system.

=====						
ERROR ANALYSIS RESULT						
=====						
Total error records: 13063						
Model	Text	Actual	Predicted	Error	Error Detail	
0 Dictionary	Social Media Managers oversee an organizations social media presence. They create and schedule content, engage with followers, and analyze social media metrics to drive brand awareness and engagement.	social media platforms (e.g., facebook, twitter, instagram) content creation and scheduling social media analytics and insights community engagement paid social advertising		False Negative	social media platforms (e.g., facebook, instagram) content creation and scheduling social media analytics and insights community engagement paid social advertising, twitter	
1 Dictionary	Frontend Web Developers design and implement user interfaces for websites, ensuring they are visually appealing and user-friendly. They collaborate with designers and backend developers to create seamless web experiences for users.	html, css, javascript frontend frameworks (e.g., react, angular) user experience (ux)		False Negative	react, css, angular) user experience (ux), javascript frontend frameworks (e.g., html	
2 Dictionary	Quality Control Managers establish and enforce quality standards within an organization. They develop quality control processes, perform inspections, and implement corrective actions to maintain product or service quality.	quality control processes and methodologies statistical process control (spc) root cause analysis and corrective action quality management systems (e.g., iso 9001) compliance and regulatory knowledge		False Negative	quality control processes and methodologies statistical process control (spc) root cause analysis and corrective action quality management systems (e.g., iso 9001) compliance and regulatory knowledge	

=====

ERROR CATEGORY SUMMARY

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	Model	Error	Count
0	Dictionary	False Negative	2000
2	Dictionary	Partial Entity	34
1	Dictionary	False Positive	28
4	Dictionary	Wrong Entity	28
3	Dictionary	Spelling Problem	7
5	ML Model	False Negative	2000
7	ML Model	Partial Entity	34
6	ML Model	False Positive	28
9	ML Model	Wrong Entity	28
8	ML Model	Spelling Problem	7
10	NER	False Negative	2000
12	NER	Partial Entity	30
11	NER	False Positive	24
13	NER	Wrong Entity	24

7	ML Model	Partial Entity	34
6	ML Model	False Positive	28
9	ML Model	Wrong Entity	28
8	ML Model	Spelling Problem	7
10	NER	False Negative	2000
12	NER	Partial Entity	30
11	NER	False Positive	24
14	NER	Wrong Entity	24
13	NER	Spelling Problem	3
15	Regex	False Negative	2000
16	Regex	False Positive	197
20	Regex	Wrong Entity	197
17	Regex	Partial Entity	161
19	Regex	Unknown Skill	105
18	Regex	Spelling Problem	104
21	TF-IDF	False Negative	2000
22	TF-IDF	False Positive	7
23	TF-IDF	Partial Entity	7
25	TF-IDF	Wrong Entity	7
24	TF-IDF	Spelling Problem	3
26	Transformer	False Negative	2000

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MODEL-WISE ERROR COUNT
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	Model	Total Errors
3	Regex	2764
0	Dictionary	2097
1	ML Model	2097
2	NER	2081
4	TF-IDF	2024
5	Transformer	2000

=====

ERROR TABLE

=====

	Model	Text	Actual	Predicted	Error
0	Dictionary	Social Media Managers oversee an organizations social media presence. They create and schedule content, engage with followers, and analyze social media metrics to drive brand awareness and engagement.	social media platforms (e.g., facebook, twitter, instagram) content creation and scheduling social media analytics and insights community engagement paid social advertising		False Negative
1	Dictionary	Frontend Web Developers design and implement user interfaces for websites, ensuring they are visually appealing and user-friendly. They collaborate with designers and backend developers to create seamless web experiences for users.	html, css, javascript frontend frameworks (e.g., react, angular) user experience (ux)		False Negative
2	Dictionary	Quality Control Managers establish and enforce quality standards within an organization. They develop quality control processes, perform inspections, and implement corrective actions to maintain product or service quality.	quality control processes and methodologies statistical process control (spc) root cause analysis and corrective action quality management systems (e.g., iso 9001) compliance and regulatory knowledge		False Negative
3	Dictionary	Wireless Network Engineers design, implement, and maintain wireless network solutions. They optimize wireless connectivity, troubleshoot issues, and ensure reliable and secure wireless communications.	wireless network design and architecture wi-fi standards and protocols rf (radio frequency) planning and optimization wireless security protocols troubleshooting wireless network issues		False Negative
4	Dictionary	A Conference Manager coordinates and manages conferences, meetings, and events. They plan logistics, handle budgeting, liaise	event planning conference logistics budget management vendor		False

False Positive

	Model	Actual	Predicted	Error Detail
98	Dictionary	sql (structured query language) database design query optimization data modeling database maintenance problem-solving skills	sql	sql
106	Dictionary	creative writing copywriting advertising copy content creation brand storytelling marketing campaigns proofreading and editing	writing	writing
124	Dictionary	backend development restful apis database integration java ee (enterprise edition) performance optimization version control (e.g., git)	java	java

False Negative

	Model	Actual	Predicted	Error Detail
0	Dictionary	social media platforms (e.g., facebook, twitter, instagram) content creation and scheduling social media analytics and insights community engagement paid social advertising	social media platforms (e.g., facebook, instagram) content creation and scheduling social media analytics and insights community engagement paid social advertising, twitter	
1	Dictionary	html, css, javascript frontend frameworks (e.g., react, angular) user experience (ux)	react, css, angular) user experience (ux), javascript frontend frameworks (e.g., html	
2	Dictionary	quality control processes and methodologies statistical process control (spc) root cause analysis and corrective action quality management systems (e.g., iso 9001) compliance and regulatory knowledge	quality control processes and methodologies statistical process control (spc) root cause analysis and corrective action quality management systems (e.g., iso 9001) compliance and regulatory knowledge	

Partial Entity

	Model	Actual	Predicted	Error Detail
100	Dictionary	sql (structured query language) database design query optimization data modeling database maintenance problem-solving skills	sql	Predicted entity partially matches actual
108	Dictionary	creative writing copywriting advertising copy content creation brand storytelling marketing campaigns proofreading and editing	writing	Predicted entity partially matches actual
126	Dictionary	backend development restful apis database integration java ee (enterprise edition) performance optimization version control (e.g.	java	Predicted entity partially matches actual

Unknown Skill

	Model	Actual	Predicted	Error Detail
2112	Regex	ui design principles and best practices graphic design tools (e.g., adobe photoshop, illustrator) typography and color theory visual design and layout responsive design	user interface	Predicted skill not present in actual annotation
2140	Regex	front-end web development html, css, javascript responsive design web performance optimization cross-browser compatibility	user experience	Predicted skill not present in actual annotation
2141	Regex	front-end web development html, css, javascript responsive design web performance optimization cross-browser compatibility	user interface	Predicted skill not present in actual annotation

Spelling Problem

	Model		Actual	Predicted	Error Detail
236	Dictionary	java programming	java frameworks (e.g.	java	Possible spelling variation
526	Dictionary	azure) devops practices automation security in the cloud disaster recovery scalability		azure	Possible spelling variation
757	Dictionary	java programming	java frameworks (e.g.	java	Possible spelling variation

DAY 17 — SKILL NORMALIZATION

1. Objective

On Day 17, I focused on **Skill Normalization** to make the extracted skills consistent and ready for real-world use.

During previous stages, the same skill could appear in different forms. For example:

Python, Python3, Python 3, Python programming

These should not be treated as four different skills. They should be standardized as:

Python

2. Why Skill Normalization is Important

Different companies and job descriptions may use different spellings, formats, or abbreviations for the same skill.

For example:

- Python3 → **Python**
- Python 3 → **Python**
- Python programming → **Python**
- PowerBI → **Power BI**
- Power-BI → **Power BI**
- Postgres → **PostgreSQL**

Without normalization, the same skill could be counted multiple times, which can affect skill frequency analysis, model evaluation, and recruitment analytics.

3. Normalization Process

Raw Extracted Skill
↓
Convert to Lowercase
↓
Remove Unnecessary Characters
↓
Match Skill Alias
↓
Apply Canonical Name
↓
Normalized Skill

Example

"python3"
↓
"python3"
↓
Alias Matching
↓
"Python"

4. Skill Alias Mapping

Raw Skill / Alias	Canonical Skill
python3	Python
python 3	Python
python programming	Python
powerbi	Power BI
power-bi	Power BI
power bi	Power BI
postgres	PostgreSQL
postgresql	PostgreSQL
ml	Machine Learning
nlp	Natural Language Processing

5. Example

Before Normalization:

Python, Python3, Python 3, PowerBI, Power-BI, Postgres

After Normalization:

Python, Power BI, PostgreSQL

This produces a cleaner and more consistent skill list.

6. Key Improvement

Before	After
Multiple names for one skill	One standard skill name
Duplicate skills	Reduced duplicates
Inconsistent spelling	Consistent naming
Difficult skill counting	Accurate skill counting
Raw extraction output	Product-ready output

7. Result

A skill normalization engine was developed to convert different skill variations and aliases into standardized names.

This makes the extracted data more reliable for skill frequency analysis, candidate-job matching, dashboards, and future machine-learning models.

Conclusion

Day 17 transformed the raw extraction results into a clean and standardized skill dataset.

The key learning was that accurate skill extraction is not enough—the extracted skills must also be normalized so that different versions of the same skill are treated as one standard skill.

	Job Title	skills	Normalized Skills
0	Digital Marketing Specialist	Social media platforms (e.g., Facebook, Twitter, Instagram) Content creation and scheduling Social media analytics and insights Community engagement Paid social advertising	Social Media Platforms, Facebook, Twitter, Instagram Content Creation And Scheduling Social Media Analytics And Insights Community Engagement Paid Social Advertising
1	Web Developer	HTML, CSS, JavaScript Frontend frameworks (e.g., React, Angular) User experience (UX)	HTML, CSS, Javascript Frontend Frameworks, React, Angular User Experience Ux
2	Operations Manager	Quality control processes and methodologies Statistical process control (SPC) Root cause analysis and corrective action Quality management systems (e.g., ISO 9001) Compliance and regulatory knowledge	Quality Control Processes And Methodologies Statistical Process Control Spc Root Cause Analysis And Corrective Action Quality Management Systems, Iso 9001 Compliance And Regulatory Knowledge
3	Network Engineer	Wireless network design and architecture Wi-Fi standards and protocols RF (Radio Frequency) planning and optimization Wireless security protocols Troubleshooting wireless network issues	Wireless Network Design And Architecture Wi Fi Standards And Protocols Rf Radio Frequency Planning And Optimization Wireless Security Protocols Troubleshooting Wireless Network Issues
4	Event Manager	Event planning Conference logistics Budget management Vendor coordination Marketing and promotion Client relations	Event Planning Conference Logistics Budget Management Vendor Coordination Marketing And Promotion Client Relations
5	Software Tester	Quality assurance processes Testing methodologies (e.g., manual, automated) Bug tracking and reporting Test case development Regression testing	Quality Assurance Processes Testing Methodologies, Manual, Automated Bug Tracking And Reporting Test Case Development Regression Testing
6	Teacher	Teaching pedagogy Classroom management Curriculum development Student assessment Differentiated instruction	Teaching Pedagogy Classroom Management Curriculum Development Student Assessment Differentiated Instruction
7	UX/UI Designer	UI design principles and best practices Graphic design tools (e.g., Adobe Photoshop, Illustrator) Typography and color theory Visual design and layout Responsive design	Ui Design Principles And Best Practices Graphic Design Tools, Adobe Photoshop, Illustrator Typography And Color Theory Visual Design And Layout Responsive Design
8	UX/UI Designer	Interaction design principles User behavior and psychology Wireframing and prototyping tools Animation and micro-interaction design Collaborative design processes	Interaction Design Principles User Behavior And Psychology Wireframing And Prototyping Tools Animation And Micro Interaction Design Collaborative Design Processes
9	Wedding Planner	Wedding planning Vendor coordination Event management	Wedding Planning Vendor Coordination Event Management

ACTUAL RAW SKILLS → CANONICAL SKILLS

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	Raw Skill	Cleaned Skill	Canonical Skill	Frequency
0	Python	python	Python	98
1	Google Analytics	google analytics	Google Analytics	86
2	Java	java	Java	80
3	CSS	css	CSS	74
4	Sketch	sketch	Sketch	73
5	AWS	aws	AWS	64
6	Interaction design principles User behavior and psychology Wireframing and prototyping tools Animation and micro-interaction design Collaborative design processes	interaction design principles user behavior and psychology wireframing and prototyping tools animation and micro interaction design collaborative design processes	Interaction Design Principles User Behavior And Psychology Wireframing And Prototyping Tools Animation And Micro Interaction Design Collaborative Design Processes	61
7	Tableau	tableau	Tableau	58
8	Illustrator) Typography and color theory Visual design and layout Responsive design	illustrator typography and color theory visual design and layout responsive design	Illustrator Typography And Color Theory Visual Design And Layout Responsive Design	53
9	Adobe Photoshop	adobe photoshop	Adobe Photoshop	53
10	UI design principles and best practices Graphic design tools (e.g.	ui design principles and best practices graphic design tools	Ui Design Principles And Best Practices Graphic Design Tools	53
11	Network management Troubleshooting Network security IT certifications (e.g.	network management troubleshooting network security it certifications	Network Management Troubleshooting Network Security It Certifications	52
12	CCNA)	ccna	Ccna	52
13	Node.js	nodejs	Node.js	49

TOP NORMALIZED SKILLS

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	Canonical Skill	Frequency
0	Python	98
1	Google Analytics	86
2	Java	80
3	CSS	74
4	Sketch	73
5	Tableau	68
6	AWS	64
7	Interaction Design Principles User Behavior And Psychology Wireframing And Prototyping Tools Animation And Micro Interaction Design Collaborative Design Processes	61
8	Adobe Photoshop	53
9	Illustrator Typography And Color Theory Visual Design And Layout Responsive Design	53
10	Ui Design Principles And Best Practices Graphic Design Tools	53
11	Ccna	52
12	Network Management Troubleshooting Network Security It Certifications	52
13	Node.js	49

